

AKADEMIKA

BEL-NBT

NETWORK AND BRIDGES TRAINER

EXPERIMENTAL MANUAL

.... A MESSAGE FROM



Today's system designers are faced with tomorrow's problems. BASIC ELECTRONICS is one of the important subject need to teach while learning electronics. It is our vision to provide you with the product you need for training ensuring lasting reliability & quality.

OUR MOTTO;

- Light years ahead – refers to leadership.

As leaders in our industry in India, We are totally committed to servicing as the standard against which all are measured in the areas of

• Design • Quality • Value • Delivery • Support.

We are truly light years ahead of our competition in this area. That means that you our valued customers are guaranteed satisfaction.

As you will move through this manual you will quickly discover that We have a complete, truly innovative & superior training products we are so committed to quality that we back our products with a complete comprehensive warranty.

AKADEMIKA LAB SOLUTIONS

Sr. No. 15/8/1, Unit No.9, Kruti Industrial Estate, Opp.

Karishma Society, Off Karve Road, Kothrud, Pune,

Maharashtra- 411038

INDIA

Contact No.:+91-7447438443

Email: lab@akademika.in

www.akademika.in

SAFETY RULES

- Read carefully and follow the instructions mentioned in this manual. This user manual includes all the important points about the installation, use and the maintenance of the product. Keep this manual always with you, for quick reference.
- After unpacking the product, arrange all the accessories in proper order, so that their integrity is checked with the packing list. Also, ensure that the accessories have no visible damage.
- Before connecting the power supply to the kit, be sure that the jumpers and the connecting chords are connected correctly, as per the experiment.
- This kit must be employed only for the use for which it has been conceived, i.e. as educational kit and must be used under the direct survey of expert personnel. Any other use is inadvisable and dangerous too. The manufacturer cannot be considered responsible for eventual damages due to improper, wrong or unreasonable uses.
- In case of any fault or malfunctioning in the trainer kit, turn off the power supply. Please do not tamper the kit. If you require our service, kindly contact the service centre for technical assistance.
- The kits are liable to malfunction/under-perform if they are not operated under standard environmental conditions of temperature and humidity.

WARRANTY

This kit is warranted against defects in workmanship and materials. Any failure due to defect in either workmanship or material should occur under normal use within a year from the original date of purchase, such failure will be corrected free of charge to the purchaser by repair or replacement of defective part or parts. When the failure is result of user's neglect, natural disaster or accident, we would charge for repairs, regardless of the warranty period. The warranty does not cover include perishable items like connecting chords, crystals, etc. and other imported items.

Conditions and Limitations

The warranty is void and inapplicable if the defective product is not brought or sent to our authorized service center or sales outlet within the warranty period. Defective product will be Akademika Lab Solutions sole judgment. The defective product will be replaced with a new one or repaired, without charge or with charge.

In the warranty period if the service is needed, the purchaser should get in touch with the service center or the sales outlet. The purchaser should return the product to the service center or the sales outlet at his or her sole expense. The loss and damage in transit will be outside the preview of this warranty. A returned product must be accompanied by a written description of the defects. Type and Model No. of the kit has to be mentioned specifically. We return the product to the purchaser at our expense. In case the warranty does not cover the product on Akademika Lab Solutions judgment, we would repair the product after obtaining prior permission from purchaser who would receive an estimate statement about the repairing charges. In such cases, Akademika Lab Solutions bares the transporting expenses required to send back all the repaired products for the moment, and then repairs and transporting expenses will be charged against the purchaser by the statement of accounts.

When the authorized sales agents sell our products, they must notify the purchaser of the warranty contents, but they have no rights to stretch the meaning of original warranty contents or to offer an additional warranty. Akademika Lab Solutions does not provide any other promise or suggestive warranty and hold no liability for the damage caused by negligence, abnormal use or natural disaster. Akademika Lab Solutions is not responsible for the damages even if it is notified of above dangers in advance as well.

For more special service or overall repairs, maintenance and up gradation of products, be sure to contact our service center or the sales outlet.

INDEX

SR. NO	CHAPTER	PAGE NO
01	Technical Specifications	06
02	Functional Block	07
03	Introduction	08
04	Experiment No 1 Study of Superposition Theorem	09
05	Experiment No.2 Study of Thevenin's Theorem	15
06	Experiment No.3 Study of Norton's Theorem	21
07	Experiment No.4 Study of Maximum Power Transfer Theorem	27
08	Experiment No.5 Study of Reciprocity Theorem	33
09	Experiment No.6 Study of Two Port Network Parameter	37
10	Experiment No.7 Study of Wheatstone Bridge	45
11	Experiment No.8 Study of Desauty Bridge	49
12	Experiment No.9 Study of Kelvin Bridge	53
13	Experiment No.10 Study of Node Voltage Analysis	57
14	Experiment No. 11 Study of Mesh Current Analysis	63
15	Experiment No. 12 Study of Source Transformation	69
16	Experiment No. 13 Study of Resistor Ladder Network	75
17	Experiment No. 14 Study of Millman's Theorem	83

18	Experiment No. 15 Study of Kirchhoff's Law	89
19	Experiment No. 16 Study of Compensation Theorem	93
20	Experiment No. 17 Study of Maxwell's Bridge	97

ASSIGNMENTS

21	Assignment No. 1 Study of Hays Bridge	103
22	Assignment No. 2 Study of Schering Bridge	107
23	Assignment No. 3 Study of Anderson Bridge	111
24	Assignment No. 4 Study of Owens Bridge	115
25	Assignment No. 5 Study of LC Resonance	119
26	Basic Component Bank	125

TECHNICAL SPECIFICATIONS

- **On Board Resources**

- Resistor Bank
- Capacitor Bank
- Potentiometers
- Gang capacitors
- Inductor bank
- Bread board

- **Power Supply**

Fixed Power Supply +25V1, -12V1, GND1, +25V2, -12V2, GND2, +25V3, -12V3, GND3

- **Bread Board**

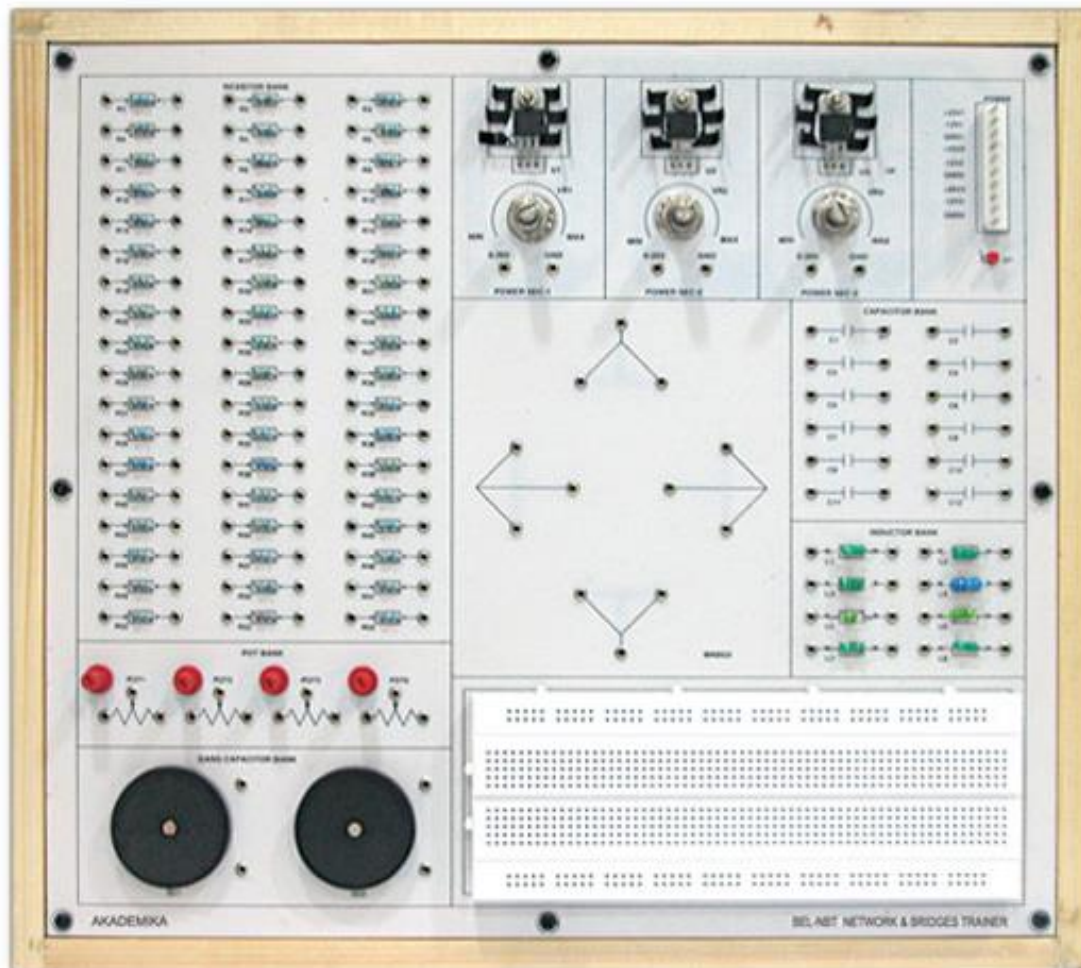
All Interconnections are made using 2mm patch cords.

Accessories

BEL-NBT			Quantity
1	Short Links PCS2 Patch cord	Male To Male	15
2	Long Links PCS2 Patch cord	Male To Male	5
3	Single Strand Wire 2m		1
4	Expt Manual		1
5	NBT Power Supply		1
6	Power cord		1

Note: Reading Tables and Graphs provided in this manual are taken on sample trainer board hence actual readings and graphs may differ from the readings and graphs shown in this manual. Readings and graphs in this manual are to be treated as sample only for reference purpose to get idea of characteristics of the respective component.

FUNCTIONAL BLOCK



INTRODUCTION

The Network and bridges Trainer is designed to cover basic theorems and ac and DC bridges. The main idea is to introduce to the students the functions and the usage of the linear ICs having pre-defined functions. Once the operation and design methodology is known, the students will be able to build applications Based on these fundamental circuits

To achieve the flexibility in design, various resources such as resistor, capacitor, diode and Potentiometer banks of different values are provided on board. Bread board area allows construction of circuits using external components along with

On-board resources

This trainer system is provided with descriptive experimental manual.

EXPERIMENT NO.1

EXPERIMENT NO.1

NAME:-

Study of Superposition Theorem

OBJECTIVE:-

To study and verify superposition theorem

EQUIPMENTS:-

- BEL-NBT Board
- Power supply
- Ammeter (0 to 100mA) - 3

THEORY:-

The **superposition theorem** for electrical circuits states that the response (Voltage or Current) in any branch of a bilateral linear circuit having more than one independent source equals the algebraic sum of the responses caused by each independent source acting alone, while all other independent sources are replaced by their internal impedances.

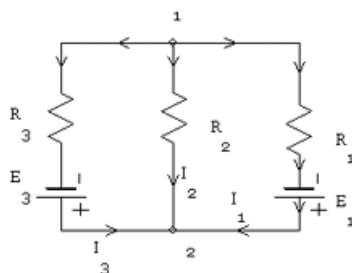
To ascertain the contribution of each individual source, all of the other sources first must be "turned off" (set to zero) by:

- Replacing all other independent voltage sources with a short circuit (thereby eliminating difference of potential. i.e. $V=0$, internal impedance of ideal voltage source is ZERO (short circuit)).
- Replacing all other independent current sources with an open circuit (thereby eliminating current. i.e. $I=0$, internal impedance of ideal current source is infinite (open circuit)).

Theoretical Calculations

The goal of following text is to check superposition theorem.

Step 1: Construct following circuit. You can also calculate currents using other techniques



Electrical scheme

Initial variables

$R_2=100\Omega$; $R_1=100\Omega$; $R_3=100\Omega$;

$E_1=3V$; $E_3=4V$;

Solution

Case I

$$V_1 \cdot G_{11} = I_{11}$$

$$G_{11} = 1/R_1 + 1/R_2 + 1/R_3 = 0.03$$

$$I_{11} = -E_1/R_1 - E_3/R_3 = -0.07$$

$$0.03 V_1 = -0.07$$

$$V_1 = -2,3333V$$

$$V_2 = 0$$

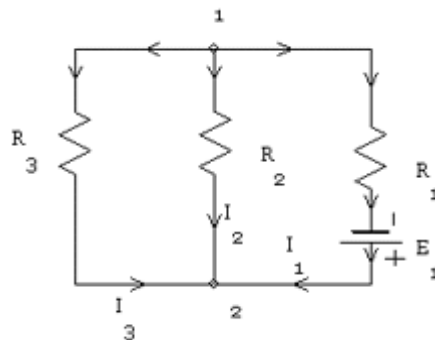
$$I_1 = (V_1 - V_2 + E_1)/R_1 = 6.7 \text{ mA}$$

$$I_2 = (V_1 - V_2)/R_2 = -23.3 \text{ mA}$$

$$I_3 = (V_1 - V_2 + E_3)/R_3 = 16.7 \text{ mA}$$

These values are used to check currents determined from superposition theorem

Step 2: Remove a voltage source from the third branch calculate the currents.



Electrical scheme

Initial variables

$R_2=100 \Omega$; $R_1=100 \Omega$; $R_3=100 \Omega$;

$E_1=3V$;

Solution

Case II

$$V_1 \cdot G_{11} = I_{11}$$

$$G_{11} = 1/R_1 + 1/R_2 + 1/R_3 = 0.03$$

$$I_{11} = -E_1/R_1 = -30 \text{ mA}$$

$$0,3V_1 = -30 \text{ mA}$$

$$V_1 = -1$$

$$V_2 = 0$$

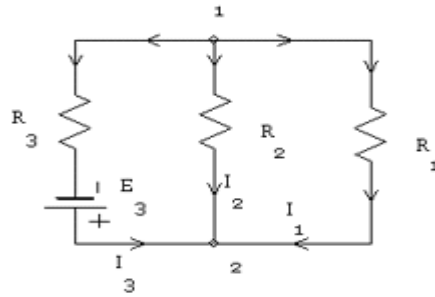
$$I_1(1) = (V_1 - V_2 + E_1)/R_1 = 20 \text{ mA}$$

$$I_2(1) = (V_1 - V_2)/R_2 = -10 \text{ mA}$$

$$I_3(1) = (V_1 - V_2)/R_3 = -10 \text{ mA}$$

These values are used to determine current from superposition theorem.

Step 3: Remove a voltage source from the first branch then run Node Voltage Analysis.



Electrical scheme

Initial variables

$R_2=10\Omega$; $R_1=10\Omega$; $R_3=10\Omega$;
 $E_3=4V$;

Solution

Case III

$$V_1 \cdot G_{11} = I_{11}$$

$$G_{11} = 1/R_1 + 1/R_2 + 1/R_3 = 0.03$$

$$I_{11} = -E_3/R_3 = -40 \text{ mA}$$

$$0,3V_1 = -40 \text{ mA}$$

$$V_1 = -1,3333V$$

$$V_2 = 0$$

$$I_1(2) = (V_1 - V_2)/R_1 = -13.3 \text{ mA}$$

$$I_2(2) = (V_1 - V_2)/R_2 = -13.3 \text{ mA}$$

$$I_3(2) = (V_1 - V_2 + E_3)/R_3 = 26.7 \text{ mA}$$

Superposition theorem checking

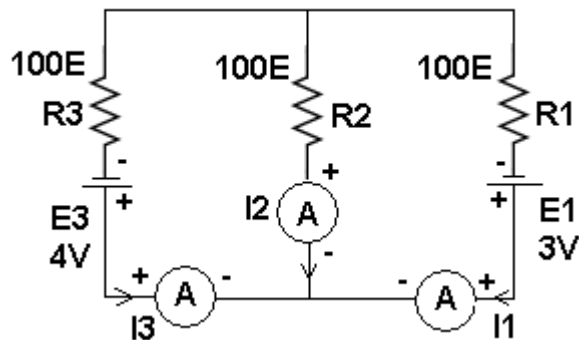
$$I_1 = I_1(1) + I_1(2) = 20 - 13.3 = 6.7 \text{ mA}$$

$$I_2 = I_2(1) + I_2(2) = -10 - 13.3 = -23.3 \text{ mA}$$

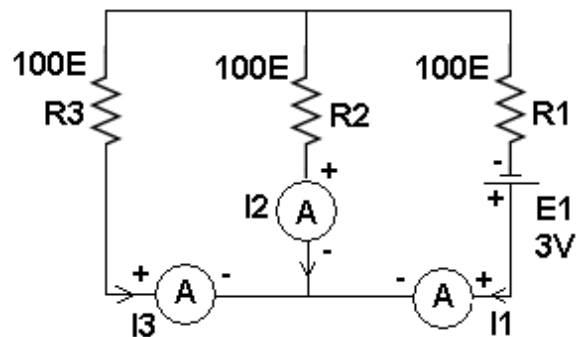
$$I_3 = I_3(1) + I_3(2) = -10 + 26.7 = 16.7 \text{ mA}$$

PROCEDURE:-

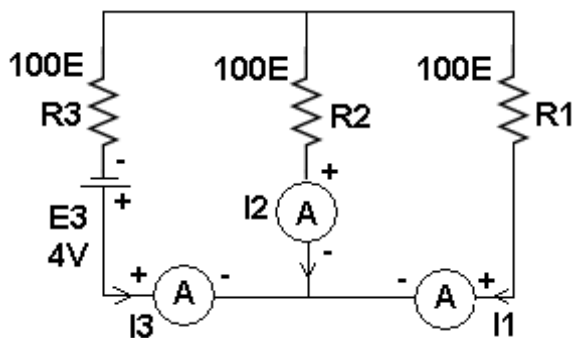
- Connect the circuit as shown.



- Make sure that the polarity of power supply and ammeter are as per diagram.
- Switch On the power supply.
- Note down the readings of currents I_1 , I_2 and I_3 from ammeter in respective branches. Tabulate the readings as **case I**
- Now remove supply E_3 from branch 3, the equivalent circuit becomes,



- Note down the readings of currents $I_{1(1)}$, $I_{2(1)}$ and $I_{3(1)}$ from ammeter and Tabulate the readings as **case II**.
- Restore the supply E_3 in branch 3 and remove the supply E_1 from branch 1 as shown below.



- Note down the readings of currents $I_{1(2)}$, $I_{2(2)}$ and $I_{3(2)}$ from ammeter and tabulate the readings as **Case III**.

OBSERVATION:-

Practically superposition theorem checking;

$$I_1 = I_1(1) + I_1(2) = 16.6 - 9.7 = 6.9$$

$$I_2 = I_2(1) + I_2(2) = -8.28 - 14.46 = -22.74$$

$$I_3 = I_3(1) + I_3(2) = -9 + 25 = 16$$

	Theoretical			Practical			
	Case I	Case II	Case III	Case I	Case II	Case III	verification
I1	6.7mA	20mA	-13.3mA	5.6mA	16.6mA	-9.7mA	6.9mA
I2	-23.3mA	-10mA	-13.3mA	-22.20mA	-8.28mA	-14.46mA	-22.74mA
I3	16.7mA	-10mA	26.7mA	17mA	-9mA	25mA	16mA

(Table given for reference only actual values may differ from above)

From above observation we can prove the super position theorem.

EXPERIMENT NO.2

EXPERIMENT NO.2

NAME:-

Study of Thevenin's Theorem

OBJECTIVE:-

To study and verify Thevenin's theorem

EQUIPMENTS:-

- BEL-NBT Board
- Power supply
- Voltmeter (0 to 20V) 1 No
- Ammeter (0 to 50mA) 1 No.

THEORY:-

In circuit theory, Thevenin's theorem for linear electrical networks states that any combination of voltage sources, current sources and resistors with two terminals is electrically equivalent to a single voltage source V and a single series resistor R . For single frequency AC systems the theorem can also be applied to general impedances, not just resistors. The theorem was first discovered by German scientist Hermann von Helmholtz in 1853[1], but was then rediscovered in 1883 by French telegraph engineer Leon Charles Thevenin (1857–1926).

His theorem states that a circuit of voltage sources and resistors can be converted into a Thevenin equivalent, which is a simplification technique used in circuit analysis. The Thevenin equivalent can be used as a good model for a power supply or battery (with the resistor representing the internal impedance and the source representing the electromotive force). The circuit consists of an ideal voltage source in series with an ideal resistor.

Thevenin's Theorem states that it is possible to simplify any linear circuit, no matter how complex, to an equivalent circuit with just a single voltage source and series resistance connected to a load. The qualification of "linear" is identical to that found in the Superposition Theorem, where all the underlying equations must be linear (no exponents or roots). If we're dealing with passive components (such as resistors, and later, inductors and capacitors), this is true. However, there are some components (especially certain gas-discharge and semiconductor components) which are nonlinear: that is, their opposition to current *changes* with voltage and/or current. As such, we would call circuits containing these types of components, nonlinear circuits.

Thevenin's Theorem is especially useful in analyzing power systems and other circuits where one particular resistor in the circuit (called the "load" resistor) is subject to change, and re-calculation of the circuit is necessary with each trial value of load resistance, to determine voltage across it and current through it.

REVIEW

- Thevenin's Theorem is a way to reduce a network to an equivalent circuit composed of a single voltage source, series resistance, and series load.
- Steps to follow for Thevenin's Theorem:
 - (1) Find the Thevenin's source voltage by removing the load resistor from the original circuit and calculating voltage across the open connection points where the load resistor used to be.
 - (2) Find the Thevenin's resistance by removing all power sources in the original circuit (voltage sources shorted and current sources open) and calculating total resistance between the open connection points.
 - (3) Draw the Thevenin's equivalent circuit, with the Thevenin's voltage source in series with the Thevenin's resistance. The load resistor re-attaches between the two open points of the equivalent circuit.
 - (4) Analyze voltage and current for the load resistor following the rules for series circuits.

PROCEDURE:-

- Take the example as shown below to verify Thevenin's theorem. calculate theoretical V_{TH} and R_{TH}

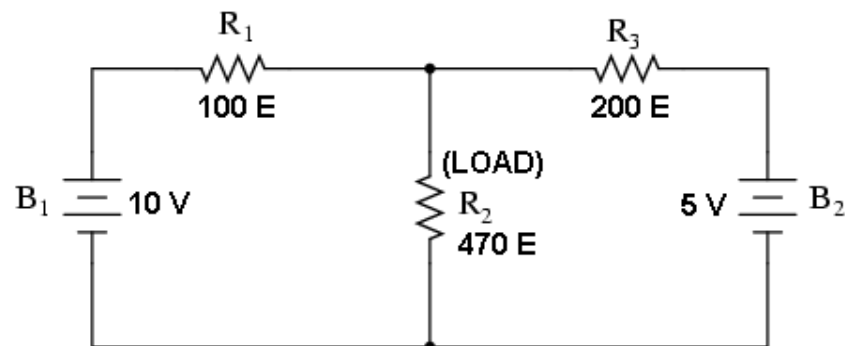
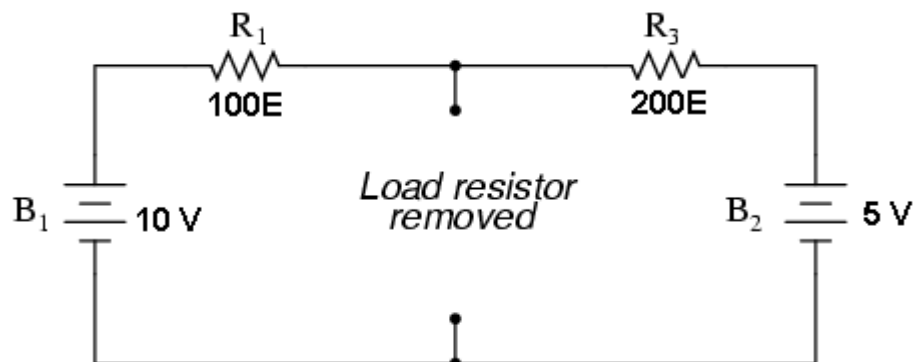


Figure 2: To verify Thevenin's theorem

(a) To Find V_{th} Theoretically

Remove load resistor and calculate the voltage appear at point where load is connected. In our case remove R_2 . The circuit will become as below.



Determine the voltage across the open load terminals by applying the rules of series circuits, Ohm's Law, and Kirchhoff's Voltage Law:

$$V = I / R$$

$$(10 - 5) = I / (100 + 200)$$

$$I = 16.67\text{mA.}$$

Thus current flowing through R1 and R3 = 16.67mA.

Now voltage across R1 and R3 can be found out by ohms law.

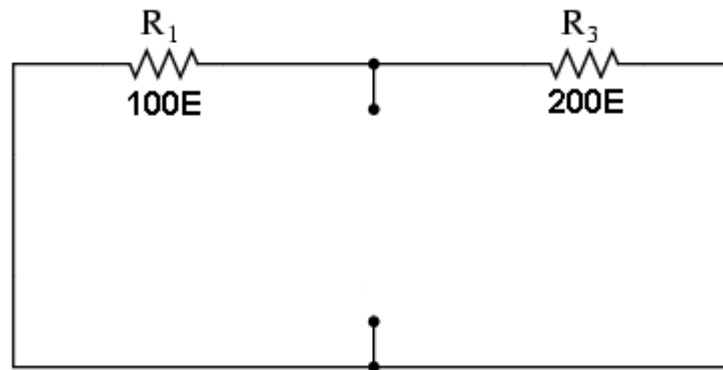
Therefore voltage across R1 = $V_{R1} = 16.67\text{mA} \times 100\Omega = 1.667\text{V}$ And voltage across R3 = $V_{R3} = 16.67\text{mA} \times 200\Omega = 3.334\text{V}$

Finally voltage across load i.e. voltage across $R_2 = V_{R2}$ is nothing but one of the battery's voltage and one of the resistor's voltage drops, and comes out to $(B1 - V_{R1})$ **8.33 volts**.

This is our "Thevenin voltage" (E_{Thevenin}) in the equivalent circuit

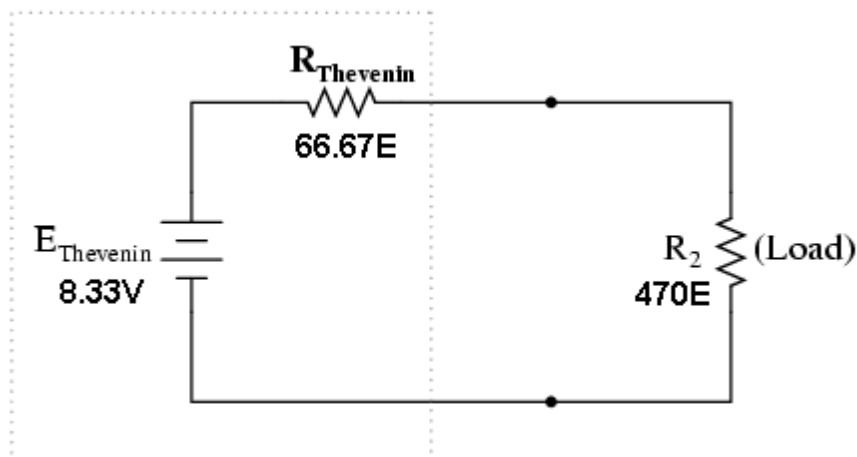
(b) To Find R_{th} Theoretically

To calculate R_{th} theoretically take original circuit and replace all voltage source by short circuit and all current source (if there) by open circuit. Remove or open load. The equivalent circuit becomes as follows.



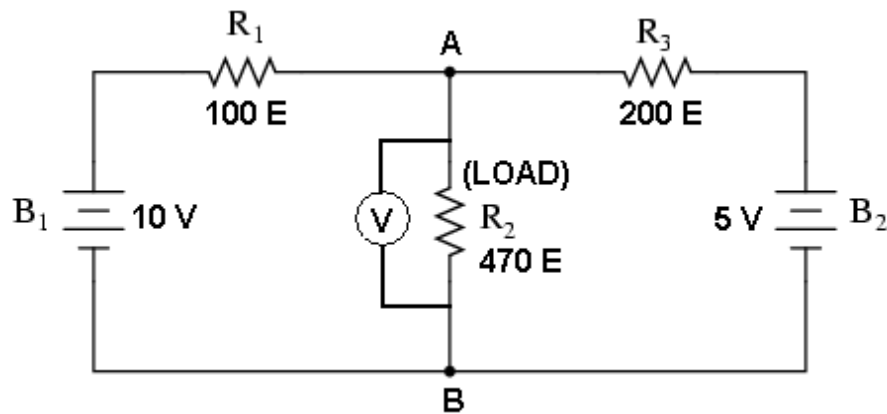
To calculate total resistance at load point take parallel combination of R1 and R3 in our case which is around = **66.67Ω**.

Now the Thevenin's Equivalent circuit will be as follows;



(c) Determine practical V_{th} and R_{th}

- Refer the original circuit and do the connection as per circuit.



- Switch OFF the supply and measure resistance at points A and B and note down this resistance as R_{TH} .
- Switch On the power supply and set B1 and B2 as per circuit diagram and measure the voltage across load and note down this voltage as V_{TH} .
- Compare the theoretical and practical values of V_{TH} and R_{TH} .

OBSERVATION:-

Verification of Thevenin's theorem.

Theoretical Observations:

	R_{TH}	R_{LOAD}	Total
V	1.11V	7.83V	8.94
I	16.67mA	16.67mA	16.67mA
R	66.67E	470E	536.67E

(Table given for reference only actual values may differ from above)

Practical Observations:

	R_{TH}	R_{LOAD}	Total
V	1.10V	7.66V	8.76V
I	16.47mA	16.47mA	16.47mA
R	66.8E	465E	531.8E

(Table given for reference only actual values may differ from above)

From the above two observations it is clear that the thevenin's theorem is verified.

EXPERIMENT NO.3

EXPERIMENT NO.3

NAME:-

Study of Norton's Theorem

OBJECTIVE:-

To study and verify Norton's Theorem

EQUIPMENTS:-

- BEL-NBT Board
- Power supply
- Voltmeter (0 to 20V) - 01
- Ammeter - 01

THEORY:-

Norton's theorem for linear electrical networks, known in Europe as the Mayer–Norton theorem, states that any collection of voltage sources, current sources, and resistors with two terminals is electrically equivalent to an ideal current source, I , in parallel with a single resistor, R . For single-frequency AC systems the theorem can also be applied to general impedances, not just resistors. The Norton equivalent is used to represent any network of linear sources and impedances, at a given frequency. The circuit consists of an ideal current source in parallel with an ideal impedance (or resistor for non-reactive circuits).

Norton's Theorem states that it is possible to simplify any linear circuit, no matter how complex, to an equivalent circuit with just a single current source and parallel resistance connected to a load. Just as with Thevenin's Theorem, the qualification of "linear" is identical to that found in the Superposition Theorem: all underlying equations must be linear (no exponents or roots).

REVIEW

- Norton's Theorem is a way to reduce a network to an equivalent circuit composed of a single current source, parallel resistance, and parallel load.
- Steps to follow for Norton's Theorem:
- Find the Norton source current by removing the load resistor from the original circuit and calculating current through a short (wire) jumping across the open connection points where the load resistor used to be.
- Find the Norton resistance by removing all power sources in the original circuit (voltage sources shorted and current sources open) and calculating total resistance between the open connection points.
- Draw the Norton equivalent circuit, with the Norton current source in parallel with the Norton resistance. The load resistor re-attaches between the two open points of the equivalent circuit.

- Analyze voltage and current for the load resistor following the rules for parallel circuits.

PROCEDURE:-

- Take the example as shown below to verify Norton's theorem. calculate theoretical I_N and R_{TH}

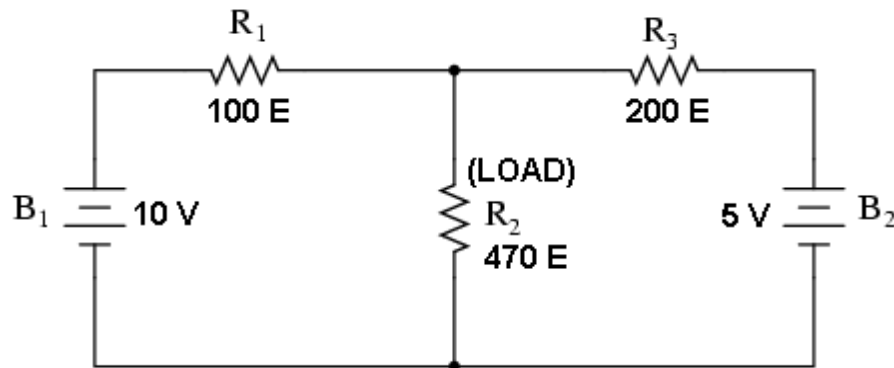
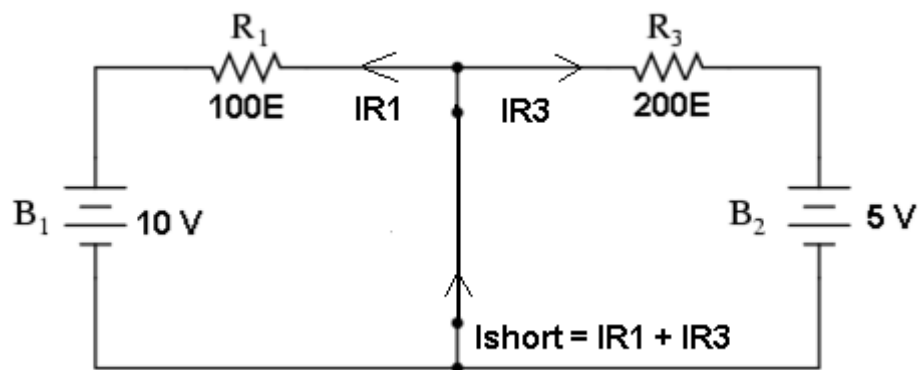


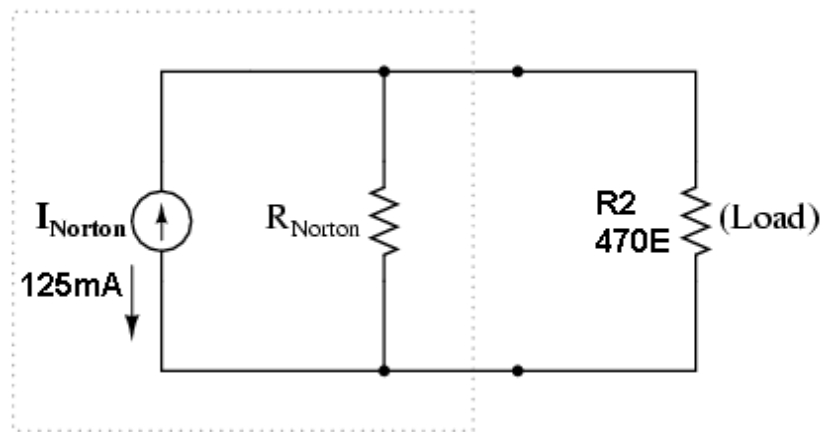
FIGURE 3: to verify Norton's Theorem

(a) To find I_N Theoretically

As before, the first step is to identify the load resistance and remove it from the original circuit. Then, to find the Norton current (for the current source in the Norton equivalent circuit), place a direct wire (short) connection between the load points and determine the resultant current. Note that this step is exactly opposite the respective step in Thevenin's Theorem, where we replaced the load resistor with a break (open circuit):

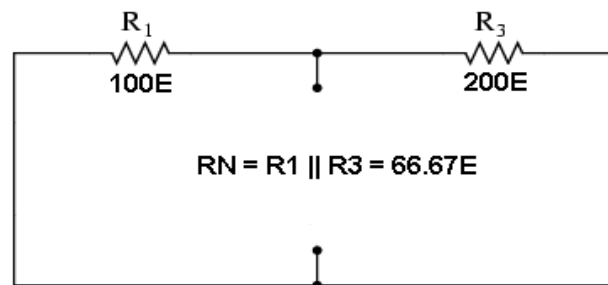


With zero voltage dropped between the load resistor connection points, the current through R_1 is strictly a function of B_1 's voltage and R_1 's resistance: 100mA ($I=E/R$). Likewise, the current through R_3 is now strictly a function of B_2 's voltage and R_3 's resistance: 25mA ($I=E/R$). The total current through the short between the load connection points is the sum of these two currents: 100mA + 25mA = 125mA. This figure of 125mA becomes the Norton source current (I_{Norton}) in our equivalent circuit:

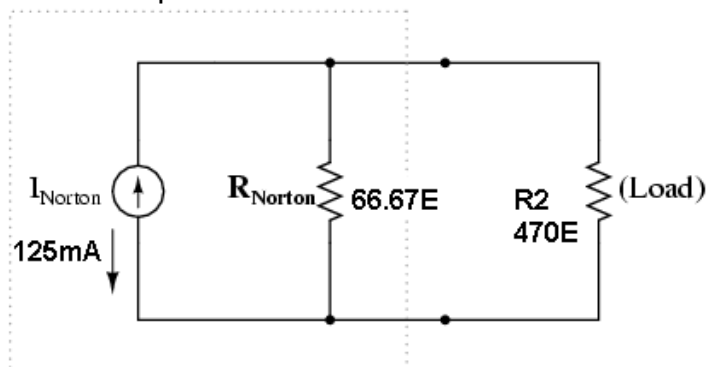


(b) To Find R_N Theoretically

To calculate the Norton resistance (R_{Norton}), we do the exact same thing as we did for calculating Thevenin resistance ($R_{Thevenin}$). Take the original circuit (with the load resistor still removed), remove the power sources (in the same style as we did with the Superposition Theorem: voltage sources replaced with wires and current sources replaced with breaks), and figure total resistance from one load connection point to the other:

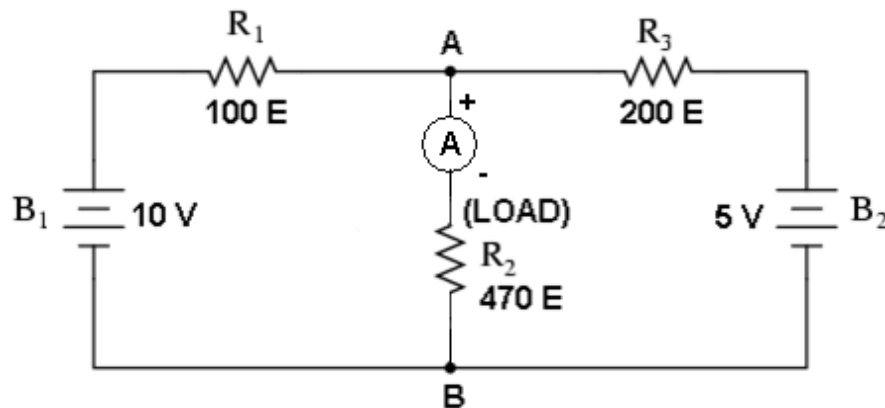


Now our Norton equivalent circuit looks like this:



(c) To find I_N & R_N Practically

- Refer the original circuit and do the connection as per circuit.



- Switch OFF the supply and measure resistance at points A and B and note down this resistance as R_N .
- Switch On the power supply and set voltage of B1 and B2 as per circuit diagram and measure the current through load and note down this voltage as I_N .
- Compare the theoretical and practical values of I_N and R_N .

OBSERVATIONS:-

Verification of Norton's theorem.

Theoretical Observations:

	R_N	R_{LOAD}	Total
V	7.3V	7.3V	7.3
I	109.5mA	15.5mA	125mA
R	66.67E	470E	58.38E

(Table given for reference only actual values may differ from above)

Practical Observations:

	R_N	R_{LOAD}	Total
V	8.3V	8.3V	8.3V
I	124mA	18mA	142mA
R	66.9E	466E	58.50E

(Table given for reference only actual values may differ from above)

From the above two observations it is clear that the Norton's theorem is verified.

EXPERIMENT NO.4

EXPERIMENT NO.4

NAME:-

Study of Maximum Power Transfer Theorem

OBJECTIVE:-

To study and verify Maximum Power Transfer Theorem

EQUIPMENTS:-

- BEL-NBT Board
- Power supply
- Voltmeter (0 to 20V) - 01
- Ammeter (0 to 200mA) - 01

THEORY:-

In electrical engineering, the **maximum power (transfer) theorem** states that, to obtain maximum external power from a source with a finite internal resistance, the resistance of the load must be made the same as that of the source. It is claimed that Moritz von Jacobi was first to discover the maximum power (transfer) theorem which is referred to as "Jacobi's law".

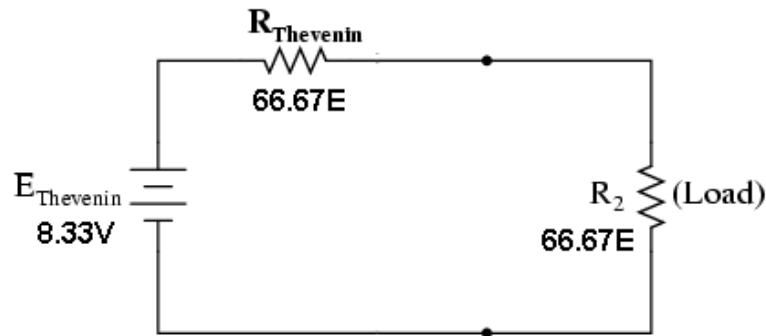
The theorem applies to maximum *power*, and not maximum efficiency. If the resistance of the load is made larger than the resistance of the source, then efficiency is higher, since most of the power is generated in the load, but the overall power is lower since the total circuit resistance goes up.

If the internal impedance is made larger than the load then most of the power ends up being dissipated in the source, and although the total power dissipated is higher, due to a lower circuit resistance, it turns out that the amount dissipated in the load is reduced.

The Maximum Power Transfer Theorem is not so much a means of analysis as it is an aid to system design. Simply stated, the maximum amount of power will be dissipated by a load resistance when that load resistance is equal to the Thevenin/Norton resistance of the network supplying the power. If the load resistance is lower or higher than the Thevenin/Norton resistance of the source network, its dissipated power will be less than maximum.

This is essentially what is aimed for in radio transmitter design, where the antenna or transmission line "impedance" is matched to final power amplifier "impedance" for maximum radio frequency power output. Impedance, the overall opposition to AC and DC current, is very similar to resistance, and must be equal between source and load for the greatest amount of power to be transferred to the load. Load impedance that is too high will result in low power output. A load impedance that is too low will not only result in low power output, but possibly overheating of the amplifier due to the power dissipated in its internal (Thevenin or Norton) impedance.

Taking our Thevenin equivalent example circuit, the Maximum Power Transfer Theorem tells us that the load resistance resulting in greatest power dissipation is equal in value to the Thevenin resistance (in this case, $66.67\ \Omega$):



The Maximum Power Transfer Theorem is not: Maximum power transfer does not coincide with maximum efficiency. Application of The Maximum Power Transfer theorem to AC power distribution will not result in maximum or even high efficiency. The goal of high efficiency is more important for AC power distribution, which dictates relatively low generator impedance compared to load impedance.

Similar to AC power distribution, high fidelity audio amplifiers are designed for relatively low output impedance and relatively high speaker load impedance. As a ratio, "output impedance" "load impedance" is known as *damping factor*, typically in the range of 100 to 1000.

Maximum power transfer does not coincide with the goal of lowest noise. For example, the low-level radio frequency amplifier between the antenna and a radio receiver is often designed for lowest possible noise. This often requires a mismatch of the amplifier input impedance to the antenna as compared with that dictated by the maximum power transfer theorem.

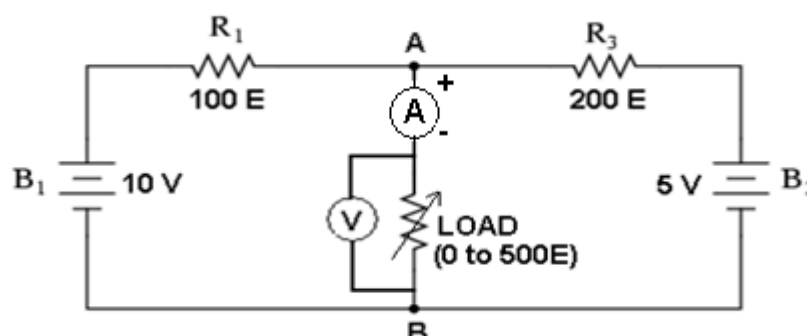
REVIEW

The Maximum Power Transfer Theorem states that the maximum amount of power will be dissipated by a load resistance if it is equal to the Thevenin or Norton resistance of the network supplying power.

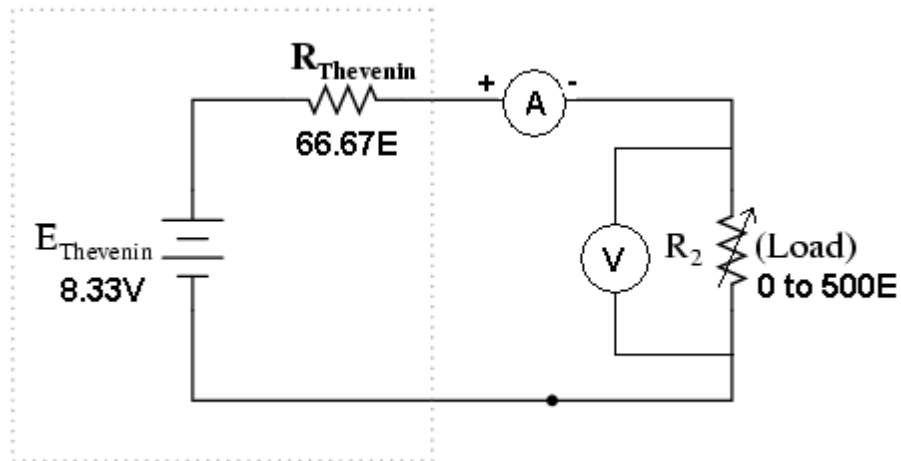
The Maximum Power Transfer Theorem does **not** satisfy the goal of maximum efficiency.

PROCEDURE:-

- Connect the circuit as shown in fig below



Thevenin's equivalent circuit of the above will be

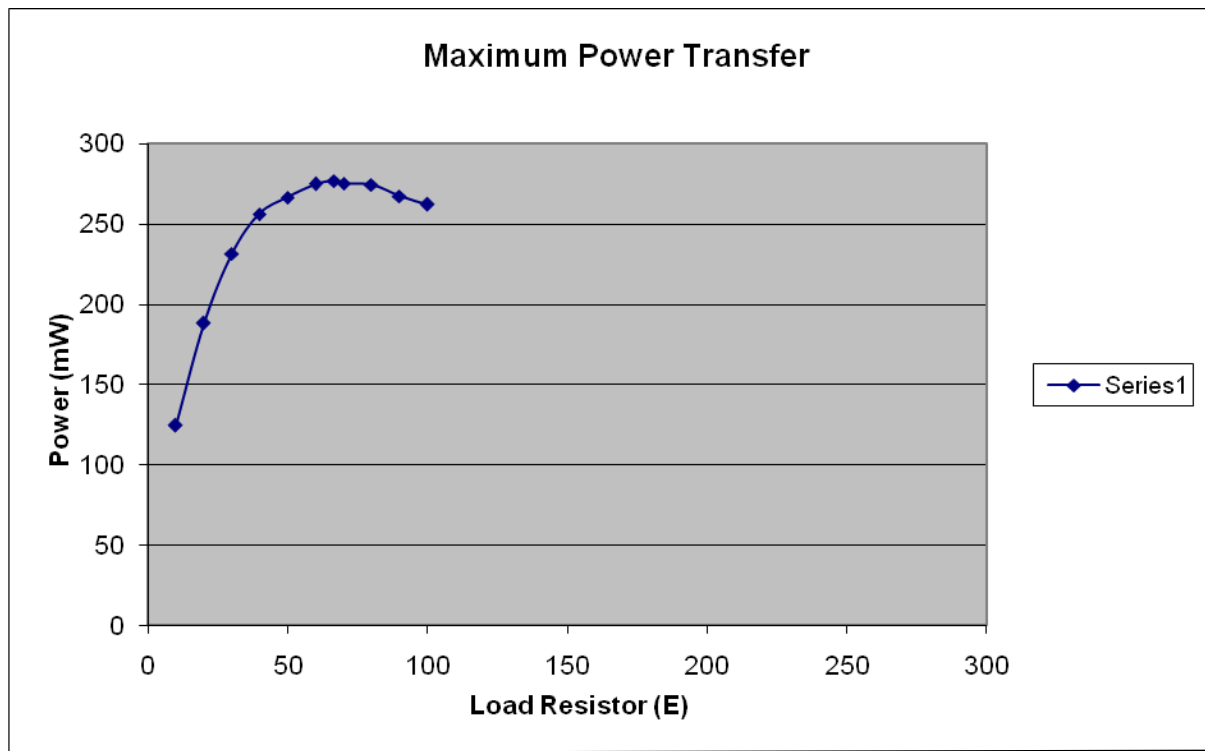


- Vary the load resistance from 10E to 100E in steps of 10 E and note down voltage across the load and current flowing through the circuit.
- Calculate power delivered to the load by using formula $P = V \times I$
- Draw the graph between resistance and power (resistance on X- axis and power on Y axis).
- Verify the maximum power is delivered to the load when $R_L = R_s$ for DC
- Plot the graph of power Vs Load resistor.

OBSERVATIONS

R	V_L volts	I_L mA	P = V_L x I_L mW
10E	1.124	111	124.32
20E	1.9	99	188.1
30E	2.6	89	231.4
40E	3.2	80	256
50E	3.6	74	266.4
60E	4.04	68	274.72
66.67E	4.26	65	276.9
70E	4.37	63	275.31
80E	4.65	59	274.35
90E	4.95	54	267.3
100E	5.14	51	262.14

(Table given for reference only actual values may differ from above)



EXPERIMENT NO.5

EXPERIMENT NO.5

NAME:-

Study of Reciprocity

OBJECTIVE:-

To study and verify reciprocity theorem

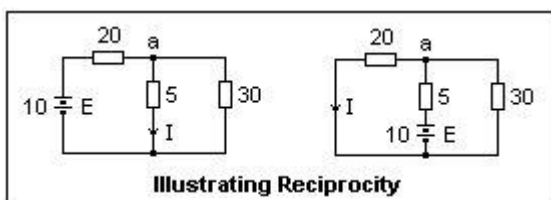
EQUIPMENTS:-

- BEL-NBT Board
- Power supply
- Ammeter (0 to 50mA) - 01

THEORY:-

In its simplest form, the reciprocity theorem states that if an emf E in one branch of a reciprocal network produces a current I in another, then if the emf E is moved from the first to the second branch, it will cause the same current in the first branch, where the emf has been replaced by a short circuit. We shall see that any network composed of linear, bilateral elements (such as R , L and C) is reciprocal.

It is sometimes phrased as the statement that voltages and currents at different points in the network can be interchanged. More technically, it follows that the mutual impedance of a first circuit due to a second is the same as the mutual impedance of the second circuit due to the first.



The circuit in the figure is a concrete example of reciprocity. The reader should solve the circuit, and determine the values of the current I in the two cases, which will be equal (0.35294 A). If E is reversed, then the direction of I is reversed, so the direction does not matter so long as both E and I are reversed at the same time. A non-bilateral element, such as a rectifying diode, destroys reciprocity. A nonlinear element also destroys reciprocity.

PROCEDURE:-

- Connect the circuit as shown in figure 5a. $R_1 = 100\Omega$, $R_2 = 180\Omega$, $R_3 = 300\Omega$, $R_4 = 220\Omega$ and $R_5 = 470\Omega$. Set 10V from power supply section and connect it with proper polarity.

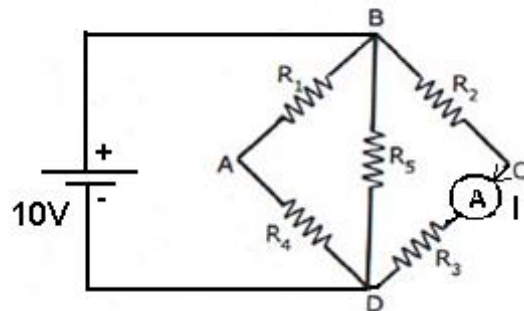


Figure 5a. Reciprocity Theorem

- Measure the current 'I' in the branch CD.
- Interchange voltage source and ammeter as shown in figure 5b and note down the current in the branch AB.

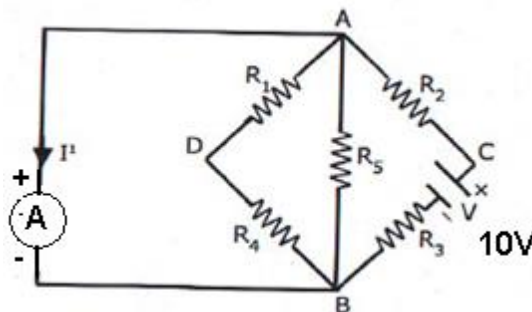


Figure 5b Reciprocity theorem.

- Observe that the current is same in both the branches AB in Figure. 5b and CD in Figure. 5a

OBSERVATION:-

SR NO.		PARAMETER	PRACTICAL VALUES
1	Before Interchange (fig 5a)	I	27mA
2.	After Interchange (fig 5b)	I	27mA

(Table given for reference only actual values may differ from above)

EXPERIMENT NO.6

EXPERIMENT NO.6

NAME:-

Study of Two Port Network Parameter

OBJECTIVE:-

- A) Study of z and Y parameter of two port network.

EQUIPMENT:-

- BEL-NBT Board
- Power supply
- Voltmeter (0 to 20V) - 02
- Ammeter (0 to 100mA) - 02

THEORY:-

Two-port network (a kind of **four-terminal network** or **quadripole**) is an electrical circuit or device with two pairs of terminals connected together internally by an electrical network. Two terminals constitute a port if they satisfy the essential requirement known as the port condition: the same current must enter and leave a port.

Examples include small-signal models for transistors (such as the hybrid-pi model), filters and matching networks. The analysis of passive two-port networks is an outgrowth of reciprocity theorems first derived by Lorentz.

A two-port network makes possible the isolation of either a complete circuit or part of it and replacing it by its characteristic parameters. Once this is done, the isolated part of the circuit becomes a "black box" with a set of distinctive properties, enabling us to abstract away its specific physical buildup, thus simplifying analysis. Any linear circuit with four terminals can be transformed into a two-port network provided that it does not contain an independent source and satisfies the port conditions.

There are a number of alternative sets of parameters that can be used to describe a linear two-port network, the usual sets are respectively called z , y , h , g , and ABCD parameters, each described individually below. These are all limited to linear networks since an underlying assumption of their derivation is that any given circuit condition is a linear superposition of various short-circuit and open circuit conditions. They are usually expressed in matrix notation, and they establish relations between the variables

$V_1 \stackrel{\text{def}}{=} \text{Input voltage}$

$V_2 \stackrel{\text{def}}{=} \text{Output voltage}$

$I_1 \stackrel{\text{def}}{=} \text{Input current}$

$I_2 \stackrel{\text{def}}{=} \text{Output current}$

which are shown in Figure 1. These current and voltage variables are most useful at low-to-moderate frequencies. At high frequencies (e.g., microwave frequencies), the use of

power and energy variables is more appropriate, and the two-port current–voltage approach is replaced by an approach based upon scattering parameters.

The terms four-terminal network and quadripole (not to be confused with quadruple) are also used, the latter particularly in more mathematical treatments although the term is becoming archaic. However, a pair of terminals can be called a *port* only if the current entering one terminal is equal to the current leaving the other; this definition is called the *port condition*. A four-terminal network can only be properly called a two-port when the terminals are connected to the external circuitry in two pairs both meeting the port condition

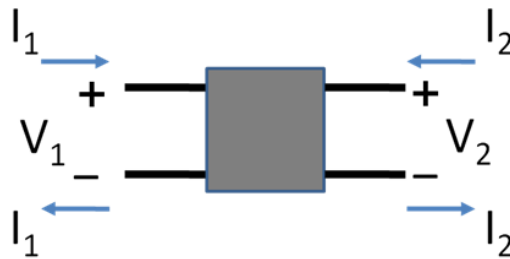
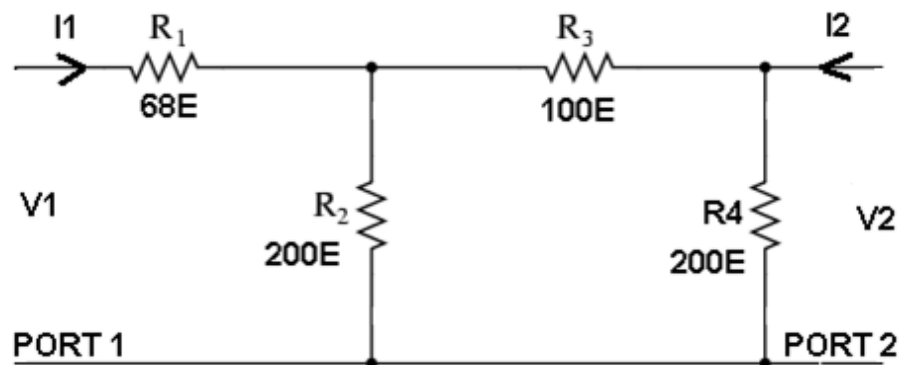


Figure 1: Example two-port network with symbol definitions. Notice the **port condition** is satisfied: the same current flows into each port as leaves that port.

Theoretical Calculations of two port network parameter



A) Z parameter

- 1) Z_{11} = it is the impedance seen looking into port 1 when port 2 is open
Therefore Z_{11} is the series and parallel combinations of the resistors R_1 to R_4 .

$$Z_{11} = R_1 + (R_2 \parallel (R_3 + R_4))$$

$$Z_{11} = 188\Omega$$

- 2) Z_{12} = it is a transfer impedance. It is the ratio of the voltage at port 1 to the current at port 2 when port 1 is open.

According to definition we need to find out values of V_1 and I_2 when port 1 is open
To find out I_2 , first find out equivalent resistance of the circuit by thevenins method. In this case $R_{TH} = (R_4 \parallel (R_2 + R_3))$

$$= 120\Omega$$

Therefore $I_2 = V_2 / R_{TH}$

$$I_2 = 5 / 120$$

$$I_2 = 41.67\text{mA}$$

To find out V_1 apply voltage divider rule,

$$V_1 = V_2 \cdot R_2 / (R_2 + R_3)$$

$$\mathbf{V_1 = 3.3V}$$

$$\text{Thus } Z_{12} = 3.3V / 41.67mA$$

$$\mathbf{Z_{12} = 79.19}$$

- 3) Z_{21} = it is a transfer impedance. It is the ratio of the voltage at port 2 to the current at port 1 when port 2 is open.

According to definition we need to find out values of V_2 and I_1 when port 1 is open

To find out I_1 , first find out equivalent resistance of the circuit by Thevenin's method.

$$R_{TH} = R_1 + (R_2 \parallel (R_3 + R_4))$$

$$\mathbf{R_{TH} = 188\Omega}$$

$$\text{Therefore } I_1 = V_1 / R_{TH}$$

$$I_1 = 10 / 188$$

$$\mathbf{I_1 = 53.19mA}$$

To find out V_2 apply voltage divider rule,

$$V_2 = V_2 \cdot R_2 / (R_2 + R_3)$$

$$\mathbf{V_1 = 3.3V}$$

$$\text{Thus } Z_{12} = 3.3V / 41.67mA$$

$$\mathbf{Z_{12} = 79.19}$$

B) Similarly Calculate Theoretically The Parameters Of Y

- 1) Y_{11} = it is the admittance seen looking into port 1 when port 2 is shorted

First we have to calculate the equivalent resistance when port 2 is shorted.

$$Y_{11} = R_1 + (R_2 \parallel (R_3))$$

$$R_{eq} = 134.67\Omega.$$

$$\text{Now } I_1 = V / R_{eq}$$

$$I_1 = 74.25mA$$

$$\text{Thus } Y_{11} = I_1 / V_1$$

$$\mathbf{Y_{11} = 7.425m}$$

- 2) Y_{12} = It is the transfer admittance. It is the ratio of the current at port 1 to the voltage at port 2 when port 1 is shorted.

First we have to calculate the equivalent resistance when port 1 is shorted to calculate the current I_1 .

$$R_{eq} = ((R_4 + R_3) \parallel R_2) + R_1$$

$$R_{eq} = 188\Omega.$$

$$\text{Thus } I_1 = 5 / 188$$

$$I_1 = 26.58mA$$

$$\text{Now } Y_{12} = I_1 / V_2$$

$$Y_{12} = 26.58mA / 5V$$

$$\mathbf{Y_{12} = 5.31m}$$

- 3) Y_{21} = It is the transfer admittance. It is the ratio of the current at port 2 to the voltage at port 1 when port 2 is shorted.

First we have to calculate the equivalent resistance when port 2 is shorted to calculate the current I_2 .

$$R_{eq} = ((R_2 + R_3) \parallel R_1)$$

$$R_{eq} = 168\Omega.$$

$$\text{Thus } I_2 = 10 / 168$$

$$I_1 = 59.52\text{mA}$$

$$\text{Now } Y_{21} = I_2 / V_1$$

$$Y_{12} = 59.52\text{mA} / 10\text{V}$$

$$Y_{21} = 5.952\text{m}$$

4) Y_{22} = it is the admittance seen looking into port 2 when port 1 is shorted. First we have to calculate the equivalent resistance when port 1 is shorted.

$$Y_{22} = ((R_1 \parallel R_2) + R_3) \parallel R_4$$

$$R_{eq} = 85.96\Omega.$$

$$\text{Now } I_2 = V / R_{eq}$$

$$I_2 = 58.17\text{mA}$$

$$\text{Thus } Y_{22} = I_2 / V_2$$

$$Y_{22} = 11.634\text{m}$$

PROCEDURE:-

Z – Parameters

- Connect the circuit as shown in figure below

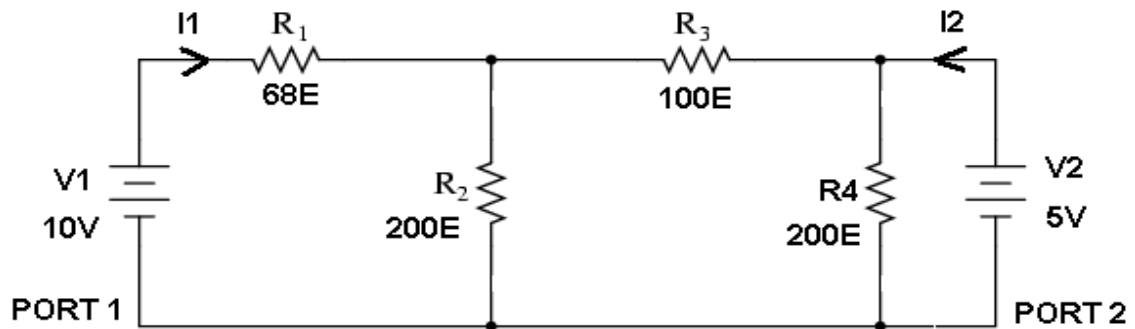
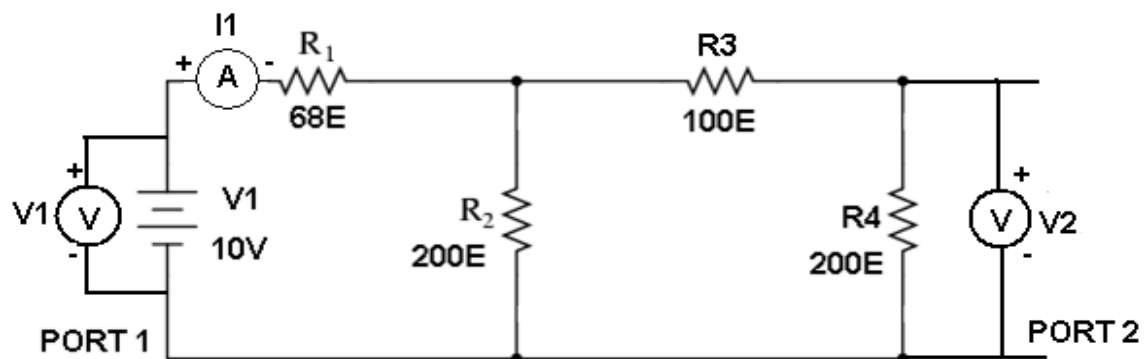


Figure 6. Z and Y parameter

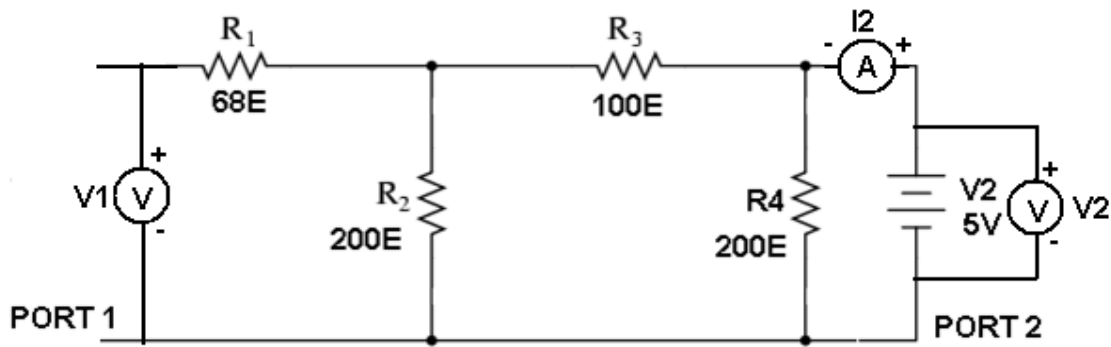
- 2. Open circuit port-2 (i.e. $I_2 = 0$) and measure V_1 , I_1 and V_2 and calculate Z_{11} & Z_{21} using the formulae



$$Z_{11} = \frac{V_1}{I_1} \mid I_2 = 0$$

$$Z_{21} = \frac{V_2}{I_1} \mid I_2 = 0$$

- To Measure Z_{12} and Z_{22} , open circuit port-1 (i.e. $I_1=0$) and measure V_1 , V_2 and I_2 and calculate Z_{12} & Z_{22} using the formulae



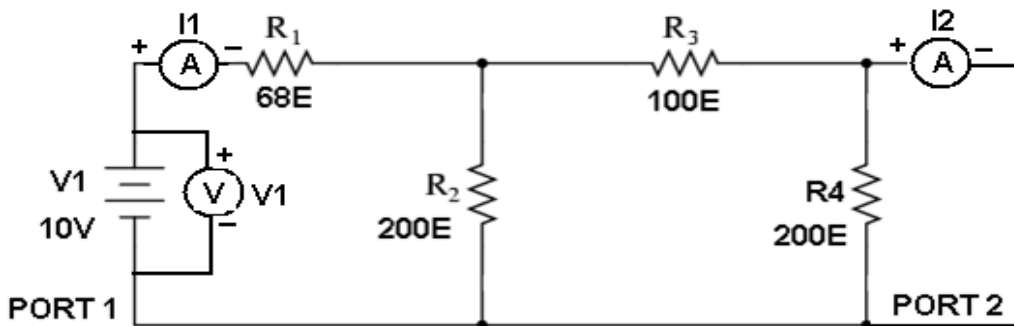
$$Z_{12} = \frac{V_1}{I_2} \mid I_1 = 0$$

$$Z_{22} = \frac{V_2}{I_2} \mid I_1 = 0$$

- Tabulate readings of Z parameter.

Y – Parameters

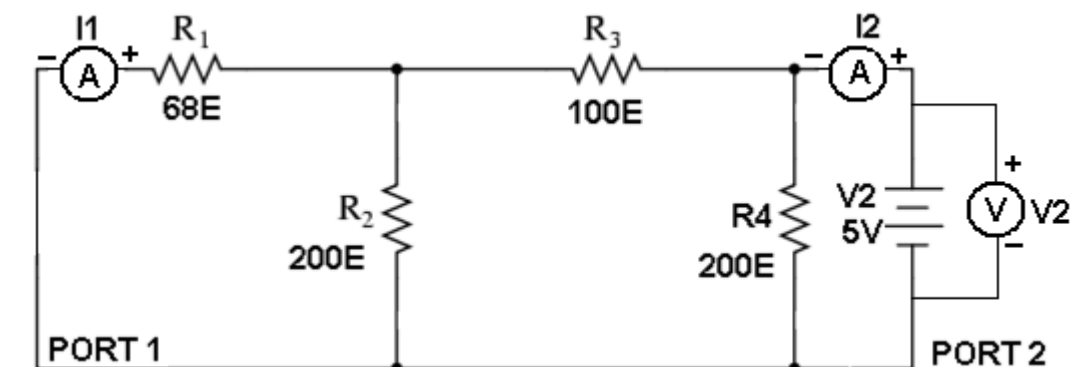
1. Connect the circuit as shown in figure 6.
2. Short circuit port-2 (i.e. $V_2 = 0$) and measure V_1 , I_1 & I_2 and calculate Y_{11} & Y_{21} using the formulae



$$Y_{11} = \frac{I_1}{V_1} \mid V_2 = 0$$

$$Y_{21} = \frac{I_2}{V_1} \mid V_2 = 0$$

- To Measure Y_{12} and Y_{22} , short circuit port-1 (i.e. $V_1=0$) and measure V_2 , I_1 and I_2 and calculate Y_{12} & Y_{22} using the formulae



$$Y_{12} = \frac{I_1}{V_2} \mid V_1 = 0$$

$$Y_{22} = \frac{I_2}{V_2} \mid V_1 = 0$$

OBSERVATION:-

Sr. No.	Parameter	Theoretical	Practical
1	Z_{11}	188	190
2	Z_{21}	120	116.28
3	Z_{12}	79.19	77.83
4	Z_{22}	120	118.4
5	Y_{11}	7.425m	7.440m
6	Y_{21}	5.952	5m
7	Y_{12}	5.31m	5.2m
8	Y_{22}	11.634m	11.683m

(Table given for reference only actual values may differ from above)

EXPERIMENT NO.7

EXPERIMENT NO.7

NAME:-

Study of Wheatstone Bridge

OBJECTIVE:-

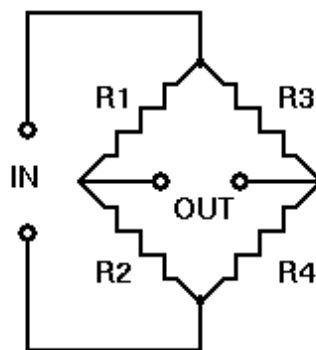
To study the working of Wheatstone bridge

EQUIPMENTS:-

- BEL-NBT Board
- Power supply
- One very sensitive dc current meter (galvanometer)

THEORY:-

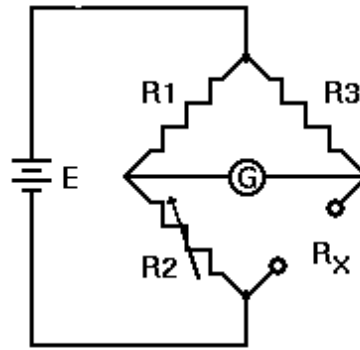
The Basic Bridge Circuit



The fundamental concept of the Wheatstone Bridge is two voltage dividers, both fed by the same input, as shown to the figure. The circuit output is taken from both voltage divider outputs.

In its classic form, a galvanometer (a very sensitive dc current meter) is connected between the output terminals, and is used to monitor the current flowing from one voltage divider to the other. If the two voltage dividers have exactly the same ratio ($R1/R2 = R3/R4$), then the bridge is said to be *balanced* and no current flows in either direction through the galvanometer. If one of the resistors changes even a little bit in value, the bridge will become unbalanced and current will flow through the galvanometer. Thus, the galvanometer becomes a very sensitive indicator of the balance condition.

Using the Wheatstone Bridge



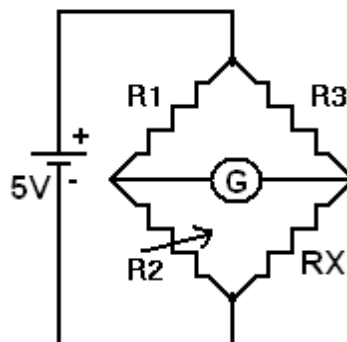
In its basic application, a dc voltage (E) is applied to the Wheatstone Bridge, and a galvanometer (G) is used to monitor the balance condition. The values of R_1 and R_3 are precisely known, but do not have to be identical. R_2 is a calibrated variable resistance, whose current value may be read from a dial or scale.

An unknown resistor, R_X , is connected as the fourth side of the circuit, and power is applied. R_2 is adjusted until the galvanometer, G , reads zero current. At this point, $R_X = R_2 \times R_3 / R_1$.

This circuit is most sensitive when all four resistors have similar resistance values. However, the circuit works quite well in any event. If R_2 can be varied over a 10:1 resistance range and R_1 is of a similar value, we can switch decade values of R_3 into and out of the circuit according to the range of value we expect from R_X . Using this method, we can accurately measure any value of R_X by moving one multiple-position switch and adjusting one precision potentiometer.

PROCEDURE:-

- Connect the circuit as shown in figure 5a.



- Connect $R_1 = 1K$ and $R_3 = 2.2K$. Connect $R_2 = 10K$ potentiometer. Connect some unknown resistance at R_X . For verification purpose connect $R_X = 4.7K$.
- Switch on the power supply and check the galvanometer pointer.
- Adjust potentiometer at R_2 till galvanometer pointer shows ZERO deflection in its scale.
- Remove the potentiometer from the bridge circuit and measure the value of the potentiometer without disturbing its shaft position.
- Calculate the value of unknown resistor R_X by formula given and verify its value.

$$R_X = R_2 (R_3 / R_1)$$
- Repeat the procedure 2 to 6 for different values of R_X and verify the results.

OBSERVATIONS:-

Sr. No.	R1	R3	R2	$R_x = R_2(R_3/R_1)$ By Calculation	Rx Theoretically
1	1K	2.2K	2.17K	4.774K	4.7K
2	1K	2.2K	2.503K	5.5K	5.6K
3	1K	2.2K	2.953	6.4966K	6.8K

(Table given for reference only actual values may differ from above)

EXPERIMENT NO.8

EXPERIMENT NO.8

NAME:-

Study of Desauty's Bridge

OBJECTIVE:-

To study the working of Desauty bridge

EQUIPMENTS:-

- BEL-NBT Board
- Power supply
- Voltmeter - 01

THEORY:-

Desauty's Bridge can be used to measure the value of a capacitor. This bridge is very simple.

Let $C_1 = C_X =$ Unknown capacitor

$C_2 =$ A Standard capacitor

R_1, R_2, R_3, R_4 are Standard Resistances.

At bridge balance condition.

$$C_1 / C_2 = R_2 / R_4 = R_1 / R_3$$

(or)

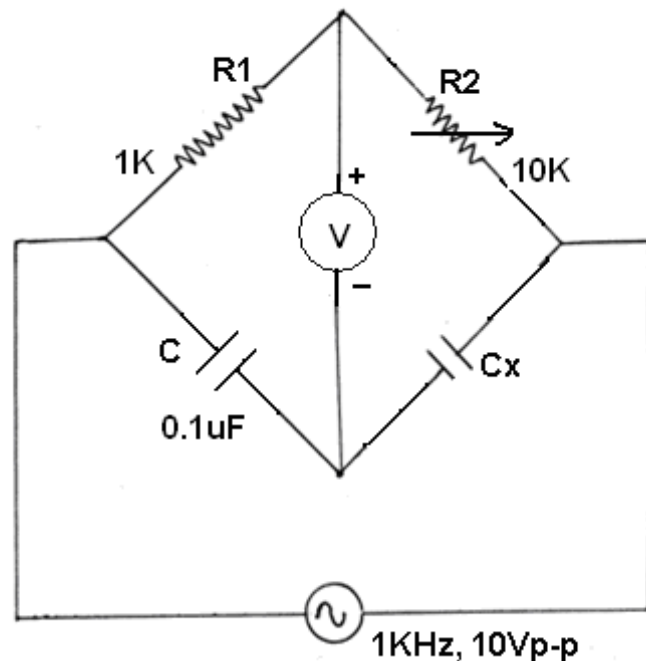
$$C_X / C_2 = R_1 / R_3 \text{ and } R_2 / R_4 = R_1 / R_3$$

Bridge balance can be obtained in two ways

- (i) Keeping C_1, C_2 Constant and varying R_1 or R_3
- (ii) Keep the values of R_1, R_2, R_3, R_4 so that $R_1/R_3 = R_2/R_4$ is satisfied and vary the capacitor C_2 . The value of unknown capacitance is found by using the formula
$$C_X = C_2 (R_1/R_3)$$

PROCEDURE:-

- Connect the circuit as shown in figure below.



- Connect $R1 = 1K$, $R2 = 10K$ potentiometer and $C = 0.1\mu F$ electrolytic capacitor. Connect some unknown capacitors at CX . For verification purpose connect $CX = 1\mu F$.
- Connect 1 KHz, 10Vp-p sine wave signal from Function Generator to bridge as shown.
- The potentiometer $R2$ is varied to obtain bridge balance condition. i.e. Voltmeter (AC mode) shows zero reading on its scale.
- Remove the potentiometer from the bridge circuit and measure the value of the potentiometer without disturbing its shaft position.
- Calculate the value of unknown resistor CX by formula given and verify its value.
 $CX = C (R1/R2)$.
- Repeat the procedure 2 to 6 for different values of CX and verify the results.

OBSERVATIONS:-

Sr. No.	R1	C	R2	$Cx = C(R1/R2)$ By Calculation	Cx Theoretically
1	1K	0.1uf	94.3E	1.06uf	1uf
2	1K	0.1uF	203.7E	0.49uF	0.47uF
3	1K	0.1uF	10.5E	9.52uF	10uF
4	1K	0.1uF	44.1E	2.26uF	2.2uF

(Table given for reference only actual values may differ from above)

EXPERIMENT NO.9

EXPERIMENT NO. 9

NAME:-

Study of Kelvin's Bridge

OBJECTIVE:-

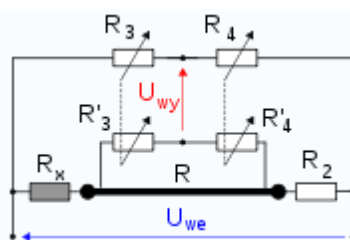
To study the working of Kelvin Bridge

EQUIPMENTS:-

- BEL-NBT Board
- Power supply
- Galvanometer - 01

THEORY:-

A Kelvin bridge (also called a Kelvin double bridge and some countries Thomson bridge) is a measuring instrument invented by William Thomson, 1st Baron Kelvin. It is used to measure an unknown low electrical resistance. Its operation is similar to the Wheatstone bridge except for the presence of additional resistors. These additional low value resistors and the internal configuration of the bridge are arranged to substantially reduce measurement errors introduced by voltage drops in the high current (low resistance) arm of the bridge



Kelvin Bridge

Principle of operation

The measurement is made by adjusting some resistors in the bridge, and the balance is achieved when:

$$R_x = R_2 \cdot \frac{R_3}{R_4} + R \cdot \frac{R_3 \cdot R'_4 - R'_3 \cdot R_4}{R_4 \cdot (R + R'_3 + R'_4)}$$

Resistance R should be as low as possible (much lower than the measured value) and for that reason is usually made as a short thick rod of solid copper. If the condition $R_3 \cdot R'_4 = R'_3 \cdot R_4$ is met (and value of R is low), then the last component in the equation can be neglected and it can be assumed that:

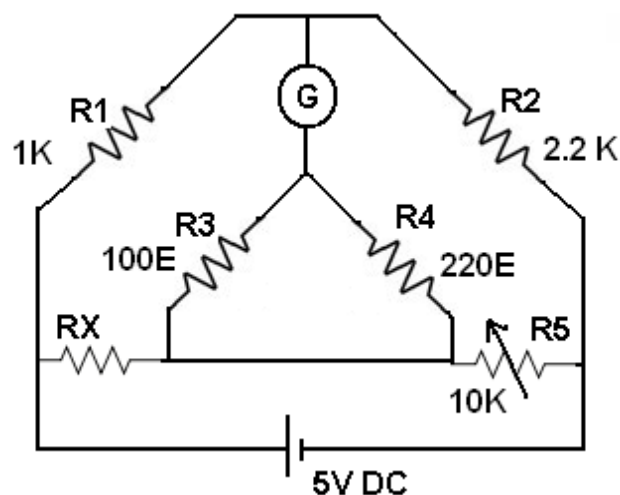
$$R_x \approx R_2 \cdot \frac{R_3}{R_4}$$

Resistance values ranging from 1 ohm to 1 mega ohm can be measured with an accuracy of approximately 0.1%. However, difficulties are encountered when very high and very low resistances are measured. Resistances less than 1 ohm are difficult to measure accurately because of uncertainty arising from the contact resistance present between the resistor to be measured and the binding posts of the bridge. Measurement of resistances greater than 1 mega ohm becomes difficult because of two factors: (1) The ratio of standard resistances R_A and R_B involve a ratio on the order of 1,000 to 1, and (2) the voltage applied to the bridge must be substantially increased to obtain definite galvanometer action. The result is that an increase in the supply voltage increases the power dissipation (heat) of the bridge resistors. The change in resistance R_B , because of the heat, is sufficient to produce an appreciable error. A Kelvin bridge is recommended for measuring resistances lower than 1 ohm. An electronic multimeter is recommended for the indicating device in bridges used for the measurement of very high resistances.

One of the most elementary precautions concerning the use of a bridge, when measuring low resistance, is to tighten the binding posts securely so that the contact resistance between the binding posts and the resistance to be measured is minimum. Leakage paths between the resistor leads along the outside surface of the resistor body must be avoided when resistances greater than 0.1 meg-ohm are measured. Search for defective solder joints or broken strands in stranded wire leads; these defects can cause erratic galvanometer indications. In those cases where wire leads must be used to reach from the resistance under test to the bridge terminals, measure the ohm value of those leads prior to further measurements.

PROCEDURE:-

- Connect the circuit as shown in figure.



- Connect $R1 = 1K$, $R2 = 2.2K$, $R3 = 100E$, $R4 = 220E$. Connect $R5 = 10K$ potentiometer. Connect some unknown resistance at R_X . For verification purpose connect $R_X = 1K$.
- Switch on the power supply and check the galvanometer pointer.
- Adjust potentiometer by $R5$ till galvanometer pointer shows ZERO deflection in its scale.
- Remove the potentiometer from the bridge circuit and measure the value of the potentiometer without disturbing its shaft position.
- Calculate the value of unknown resistor R_X by formula given and verify its value.

- $R_X = R_5 (R_1/R_2)$.
- Repeat the procedure 2 to 6 for different values of R_X and verify the results.

OBSERVATIONS:-

Sr. No.	R1	R2	R3	R4	R5	$R_X = R_5(R_1/R_2)$ By Calculation	R_X Theoretically
1	1K	2.2K	100E	220E	2.218K	1.008K	1K
2	1K	2.2K	100E	220E	4.88K	2.218K	2.2K
3	1K	2.2K	100E	220E	490E	222.7	220E

(Table given for reference only actual values may differ from above)

For very low value resistance measurement decrease the ratio of resistance R_1/R_2 and R_2/R_3 .

Sr. No.	R1	R2	R3	R4	R5	$R_X = R_5(R_1/R_2)$ By Calculation	R_X Theoretically
1	1K	10K	100E	1K	57E	5.7E	4.7E
2	1K	10K	100E	1K	97.9E	9.8E	10E
3	1K	10K	100E	1K	492E	49.2E	47E

(Table given for reference only actual values may differ from above)

EXPERIMENT NO.10

EXPERIMENT NO. 10

NAME:-

Study of Node Voltages

OBJECTIVE:-

To study the Node voltage analysis and verify it practically

EQUIPMENTS:-

- BEL-NBT Board
- Power supply
- Voltmeter - 02

THEORY:-

In electric circuits analysis, nodal analysis, node-voltage analysis, or the branch current method is a method of determining the voltage (potential difference) between "nodes" (points where elements or branches connect) in an electrical circuit in terms of the branch currents.

In analyzing a circuit using Kirchhoff's circuit laws, one can either do nodal analysis using Kirchhoff's current law (KCL) or mesh analysis using Kirchhoff's voltage law (KVL). Nodal analysis writes an equation at each electrical node, requiring that the branch currents incident at a node must sum to zero. The branch currents are written in terms of the circuit node voltages. As a consequence, each branch constitutive relation must give current as a function of voltage; an admittance representation. For instance, for a resistor, $I_{\text{branch}} = V_{\text{branch}} * G$, where $G (=1/R)$ is the admittance of the resistor.

Nodal analysis is possible when all the circuit elements' branch constitutive relations have an admittance representation. Nodal analysis produces a compact set of equations for the network, which can be solved by hand if small, or can be quickly solved using linear algebra by computer. Because of the compact system of equations, many circuit simulation programs (e.g. SPICE) use nodal analysis as a basis. When elements do not have admittance representations, a more general extension of nodal analysis, modified nodal analysis, can be used.

While simple examples of nodal analysis focus on linear elements, more complex nonlinear networks can also be solved with nodal analysis by using Newton's method to turn the nonlinear problem into a sequence of linear problems.

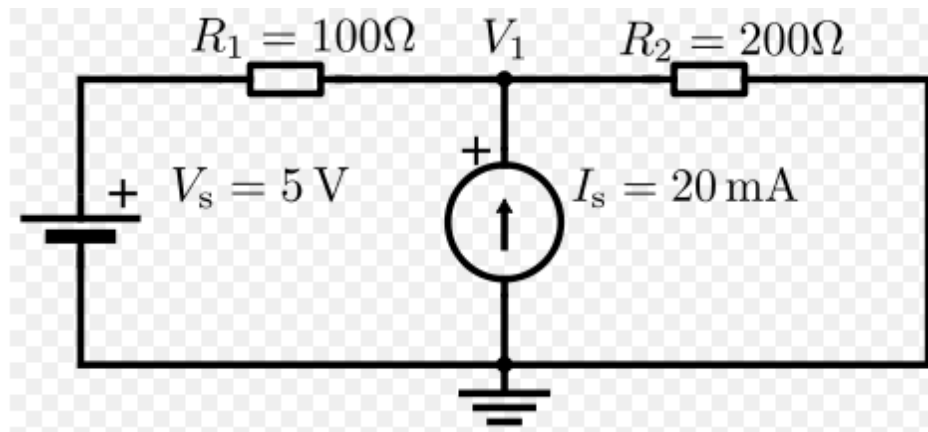
Method

- Note all connected wire segments in the circuit. These are the *nodes* of nodal analysis.

- Select one node as the ground reference. The choice does not affect the result and is just a matter of convention. Choosing the node with most connections can simplify the analysis.
- Assign a variable for each node whose voltage is unknown. If the voltage is already known, it is not necessary to assign a variable.
- For each unknown voltage, form an equation based on Kirchhoff's current law. Basically, add together all currents leaving from the node and mark the sum equal to zero.
- If there are voltage sources between two unknown voltages, join the two nodes as a super node. The currents of the two nodes are combined in a single equation, and a new equation for the voltages is formed.
- Solve the system of simultaneous equations for each unknown voltage.

Basic example

The only unknown voltage in this circuit is V_1 . There are three connections to this node and consequently three currents to consider. The direction of the currents in calculations is chosen to be away from the node.



- Current through resistor R_1 : $(V_1 - V_S) / R_1$
- Current through resistor R_2 : V_1 / R_2
- Current through current source I_S : $-I_S$

With Kirchhoff's current law, we get:

$$\frac{V_1 - V_S}{R_1} + \frac{V_1}{R_2} - I_S = 0$$

This equation can be solved in respect to V_1 :

$$V_1 = \left(\frac{V_S}{R_1} + I_S \right) : \left(\frac{1}{R_1} + \frac{1}{R_2} \right)$$

Finally, the unknown voltage can be solved by substituting numerical values for the symbols. Any unknown currents are easy to calculate after all the voltages in the circuit are known.

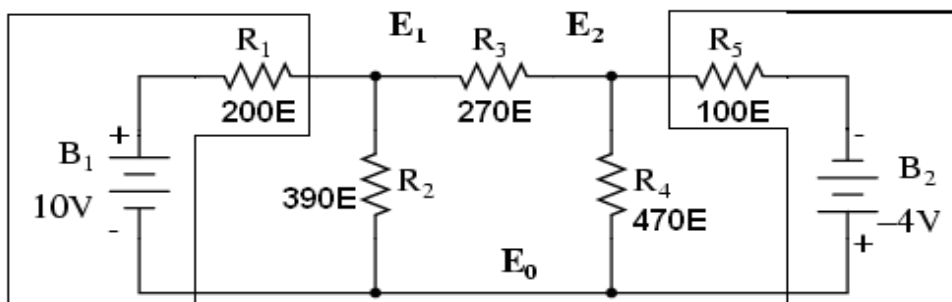
$$V_1 = \left(\frac{5 \text{ V}}{100 \Omega} + 20 \text{ mA} \right) : \left(\frac{1}{100 \Omega} + \frac{1}{200 \Omega} \right) \approx 4.667 \text{ V}$$

PROCEDURE:-

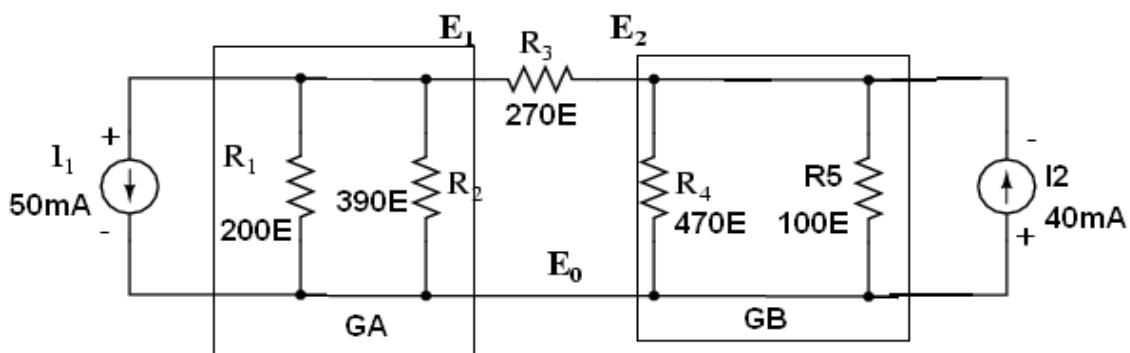
Theoretical Analysis of Node Voltage

The node voltage method of analysis solves for unknown voltages at circuit nodes in terms of a system of KCL equations. This analysis looks strange because it involves replacing voltage sources with equivalent current sources. Also, resistor values in ohms are replaced by equivalent conductance's in Siemens, $G = 1/R$. The Siemens (S) is the unit of conductance, having replaced the mho unit. In any event $S = \Omega^{-1}$. And $S = \text{mho}$ (obsolete).

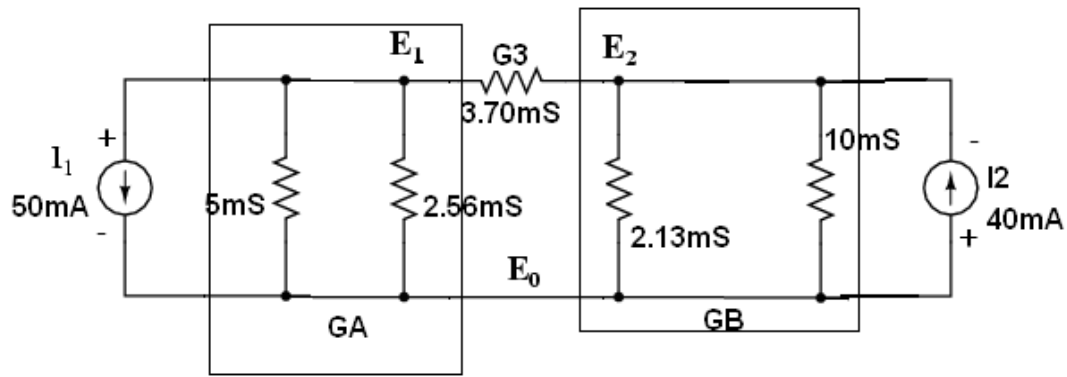
We start with a circuit having conventional voltage sources. A common node E_0 is chosen as a reference point. The node voltages E_1 and E_2 are calculated with respect to this point.



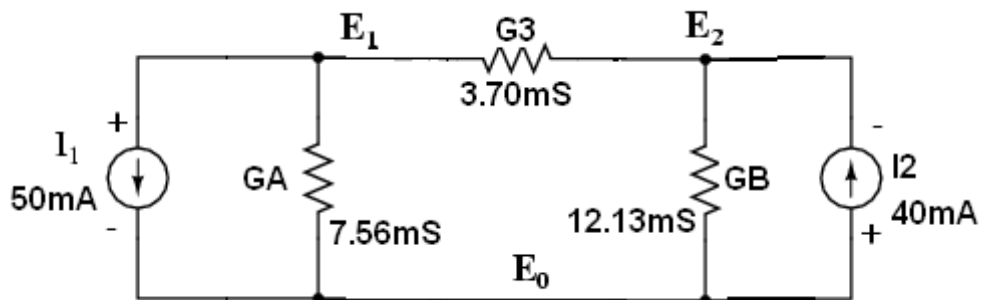
- Replace voltage source from above circuit to current source.



- Substitute resistances by their equivalent conductance in Siemens $G = 1/R$.



- Replace G_A and G_B by their equivalent parallel values by addition.



- Deriving a general node voltage method, we write a pair of KCL equations in terms of unknown node voltages E_1 and E_2 this one time. We do this to illustrate a pattern for writing equations by inspection

$$G_A E_1 + G_3 (E_1 - E_2) = I_1 \quad (1)$$

$$G_B E_2 - G_3 (E_1 - E_2) = I_2 \quad (2)$$

Simplifying the equation 1 & 2 we get

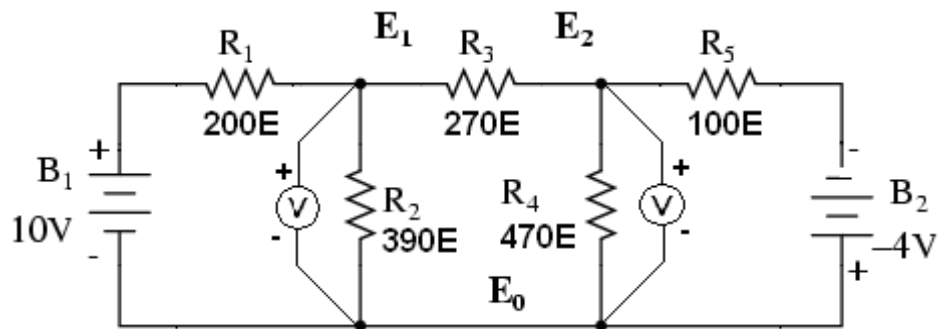
$$(G_A + G_3) E_1 - G_3 E_2 = I_1 \quad (3)$$

$$-G_3 E_1 + (G_B + G_3) E_2 = I_2 \quad (4)$$

- The coefficients of the last pair of equations above have been rearranged to show a pattern. The sum of conductances connected to the first node is the positive coefficient of the first voltage in equation (3). The sum of conductance connected to the second node is the positive coefficient of the second voltage in equation (4). The other coefficients are negative, representing conductances between nodes. For both equations, the right hand side is equal to the respective current source connected to the node. This pattern allows us to quickly write the equations by inspection. This leads to a set of rules for the node voltage method of analysis.
- Solving the simultaneous equation 3 & 4 we get **$E_1 = 4.01V$ and $E_2 = -1.40V$** .

Practical verification of Node Voltage Analysis

- Do the connection as shown in circuit below.



- Set $B_1 = 10V$ and $B_2 = -4V$ and connect them into circuit with proper polarity.
- Switch on the power supply and measure the voltages E_1 and E_2 with respective voltmeter.
- Note down the readings and compare the theoretical and practical readings.

OBSERVATIONS:-

SR NO	Voltages (theoretical)	Voltages (Practical)
1	$E_1 = 4.01V$	$E_1 = 3.275V$
2	$E_2 = -1.40V$	$E_2 = -1.79V$

(Table given for reference only actual values may differ from above)

EXPERIMENT NO.11

EXPERIMENT NO. 11

NAME:-

Study of Mesh Current

OBJECTIVE:-

To study mesh current analysis and verify it practically

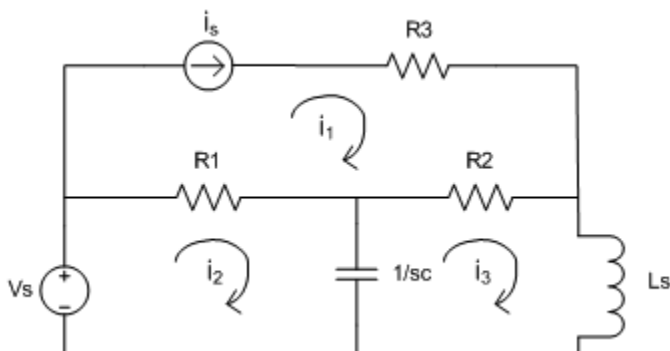
EQUIPMENTS:-

- BEL-NBT Board
- Power supply
- Ammeter - 2

THEORY:-

Mesh analysis (sometimes referred to as loop analysis or mesh current method) is a method that is used to solve planar circuits for the voltage and currents at any place in the circuit. Planar circuits are circuits that can be drawn on a plane with no wires overlapping each other. Mesh analysis uses Kirchhoff's voltage law to solve these planar circuits. The advantage of using mesh analysis is that it creates a systematic approach to solving planar circuits and reduces the number of equations needed to solve the circuit for all of the voltages and currents.

Mesh analysis works by arbitrarily assigning mesh currents in the essential meshes. An essential mesh is a loop in the circuit that does not contain any other loop. When looking at a circuit schematic, the essential meshes look like a "window pane". Figure below labels the essential meshes with one, two, and three. Once the essential meshes are found, the mesh currents need to be labeled.



A mesh current is a current that loops around the essential mesh. The mesh current might not have a physical meaning but it is used to set up the mesh analysis equations. When assigning the mesh currents it is important to have all the mesh currents loop in the same direction. This will help prevent errors when writing out the equations. The convention is to have all the mesh currents looping in a clockwise direction.

The reason to use mesh currents instead of just using KCL and KVL to solve a problem is that the mesh currents can account for any unnecessary currents that may be drawn in when using KCL and KVL. Mesh analysis ensures that the least possible number of equations regarding currents is used, greatly simplifying the problem.

After labeling the mesh currents, one only needs to write one equation per mesh in order to solve for all the currents in the circuit. These equations are the sum of the voltage drops in a complete loop of the mesh current.^[2] For other than current and voltage sources, the voltage drops will be the impedance of the electronic component multiplied by the mesh current in that loop. It is important to note that if a component exists between two essential meshes, the component's voltage drop will be the impedance of the component times the present mesh current minus the neighboring mesh current (computing the subtraction first).

If a voltage source is present within the mesh loop, the voltage at the source is either added or subtracted depending on if it is a voltage drop or a voltage rise in the direction of the mesh current. For a current source that is not contained between two meshes, the mesh current will take the positive or negative value of the current source depending on if the mesh current is in the same or opposite direction of the current source.^[2] The following is the same circuit from above with the equations needed to solve for all the currents in the circuit.

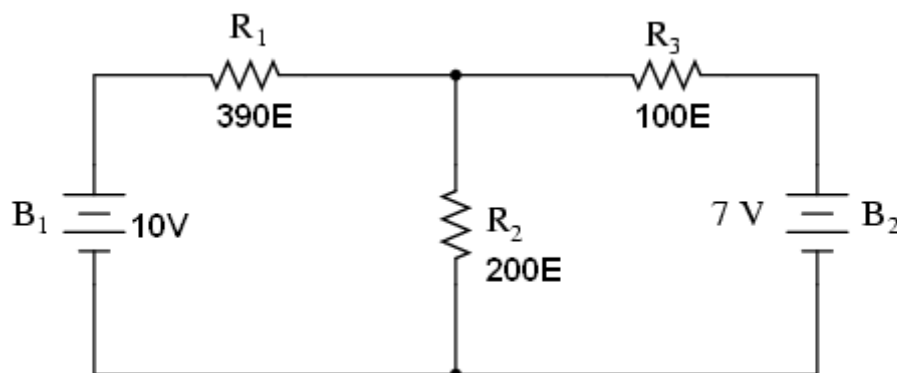
$$\begin{cases} \text{Mesh 1: } i_1 = i_s \\ \text{Mesh 2: } -V_s + R_1(i_2 - i_1) + \frac{1}{s_c}(i_2 - i_3) = 0 \\ \text{Mesh 3: } \frac{1}{s_c}(i_3 - i_2) + R_2(i_3 - i_1) + Lsi_3 = 0 \end{cases}$$

Once the equations are found, the system of linear equations can be solved by using any technique to solve linear equations.

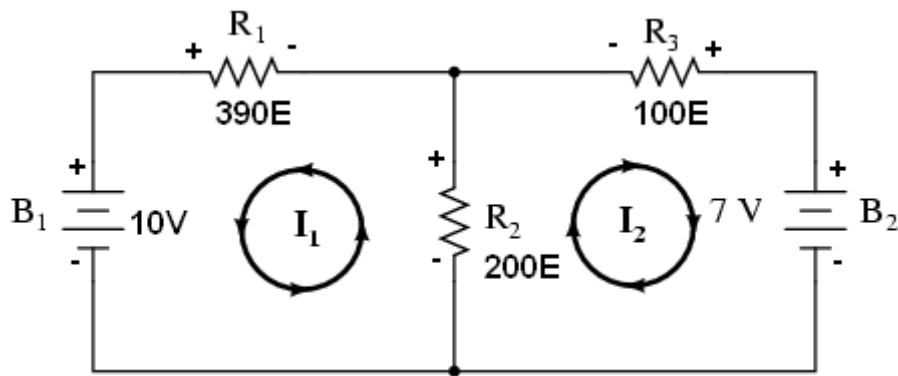
PROCEDURE:-

Theoretical Analysis of Mesh Current

Let's see how this method works on the example below



- Identify the loops and assume current direction.



- Tracing the left loop of the circuit, starting from the upper-left corner and moving counter-clockwise (the choice of starting points and directions is ultimately irrelevant), counting polarity as if we had a voltmeter in hand, red lead on the point ahead and black lead on the point behind, we get this equation:

$$-10 + (I_1 + I_2) 200 + 390 (I_1) = 0$$

Simplifying the equation

$$-10 + 590 I_1 + 200 I_2 = 0 \quad (1)$$

- At this time we have one equation with two unknowns. To be able to solve for two unknown mesh currents, we must have two equations. If we trace the other loop of the circuit, we can obtain another KVL equation and have enough data to solve for the two currents. Creature of habit that I am, I'll start at the upper-left hand corner of the right loop and trace counter-clockwise:

$$-200 (I_1 + I_2) + 7 - 100 I_2 = 0$$

Simplifying the equation

$$-200 I_1 - 300 I_2 = -7 \quad (2)$$

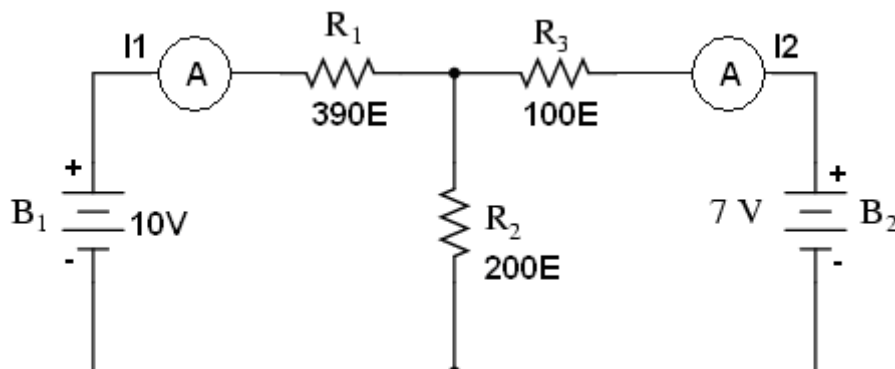
Solving the simultaneous equation 1 & 2 values of I_1 and I_2 can be calculated,

$$I_1 = 11.46\text{mA}$$

$$I_2 = 16.18\text{mA}$$

Practical verification of mesh current Analysis;

- Do the connection as shown in circuit below.



- Set $B_1 = 10\text{V}$ and $B_2 = 7\text{V}$ and connect them into circuit with proper polarity.
- Switch on the power supply and measure the currents I_1 and I_2 with respective ammeter.
- Note down the readings and compare the theoretical and practical readings.

OBSERVATIONS:-

SR NO	Current (theoretical)	Current (Practical)
1	I1 = 11.46mA	I1 = 14mA
2	I2 = 16.18mA	I2 = 13.1mA

(Table given for reference only actual values may differ from above)

EXPERIMENT NO.12

EXPERIMENT NO. 12

NAME:-

Study of Source Transformation

OBJECTIVE:-

To study the concept of source transformation

EQUIPMENTS:-

- BEL-NBT Board
- Power supply
- Voltmeter - 2

THEORY:-

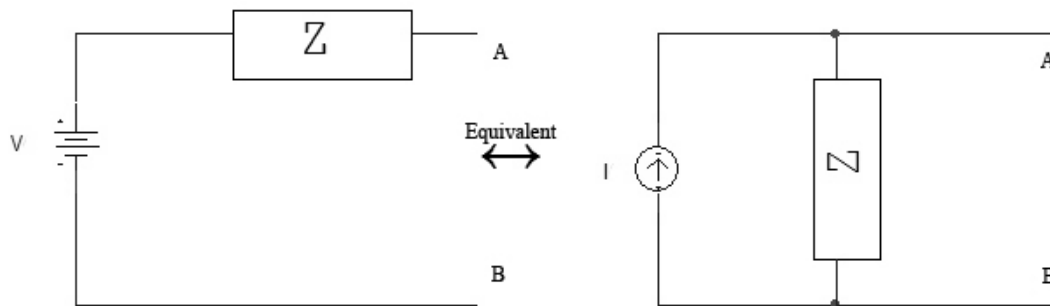
Finding a solution to a circuit can be somewhat difficult without using tricks or methods that make the circuit appear simpler. Circuit solutions are often simplified, especially with mixed sources, by transforming a voltage into a current source, and vice versa. This process is known as a **source transformation**, and is an application of Thevenin's theorem and Norton's theorem.

Performing a source transformation is the process of using Ohms Law to take an existing voltage source in series with a resistance, and replace it with a current source in parallel with the same resistance. Remember that Ohms law states that a voltage in a material is equal to the material's resistance times the amount of current through it. Since source transformations are bilateral, one can be derived from the other. ^[2] Source transformations are not limited to resistive circuits however. They can be performed on a circuit involving capacitors and inductors, as long as the circuit is first put into the frequency domain. In general, the concept of source transformation is an application of Thevenin's theorem to a current source, or Norton's theorem to a voltage source.

Specifically, source transformations are used to exploit the equivalence of a real current source and a real voltage source, such as a battery. Application of Thevenin's theorem and Norton's theorem gives the quantities associated with the equivalence. Specifically, suppose we have a real current source I , which is an ideal current source in parallel with impedance. If the ideal current source is rated at I amperes, and the parallel resistor has an impedance Z , then applying a source transformation gives an equivalent real voltage source, which is ideal, and in series with the impedance. This new voltage source V , has a value equal to the ideal current source's value times the resistance contained in the real current source. The impedance component of the real voltage source retains its real current source value

Source transformations are easy to perform as long as there is a familiarity with Ohms Law. Basically, if there is a voltage source in series with an impedance, It is possible to find the value of the equivalent current source in parallel with the impedance by dividing the value of the voltage source by the value of the impedance. The converse also applies

here. Initially, if a current source in parallel with impedance is present, multiplying the value of the current source with the value of the impedance will result in the equivalent voltage source in series with the impedance. A visual example of what is being done during a source transformation can be seen in Figure below

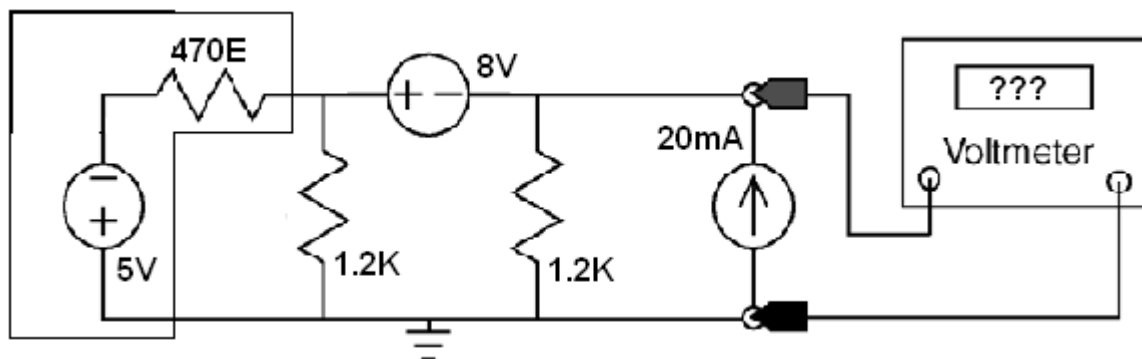


An example of a DC source transformation. Notice that the impedance Z is the same in both configurations

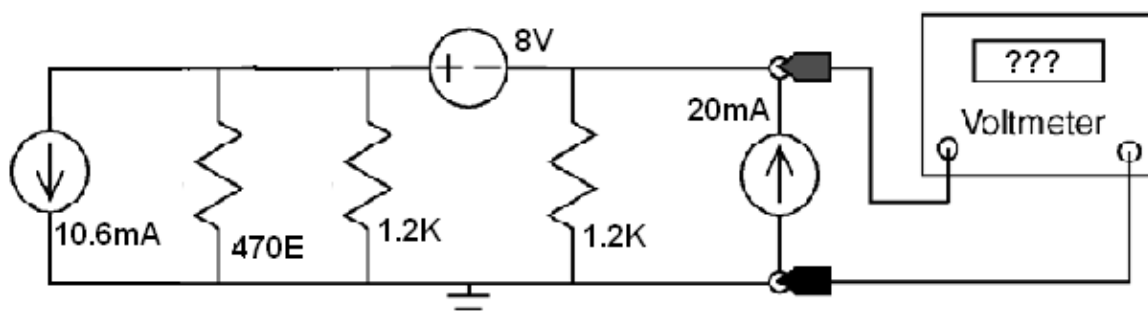
PROCEDURE:-

Theoretical Analysis of Source Transformation

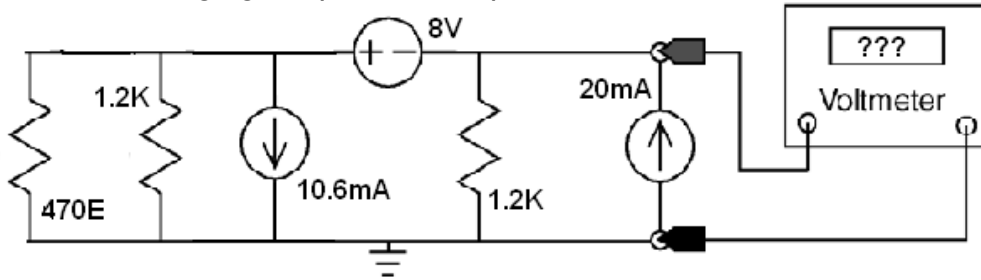
Consider the circuit shown in Figure below. Find the value of the voltage measured by the voltmeter.



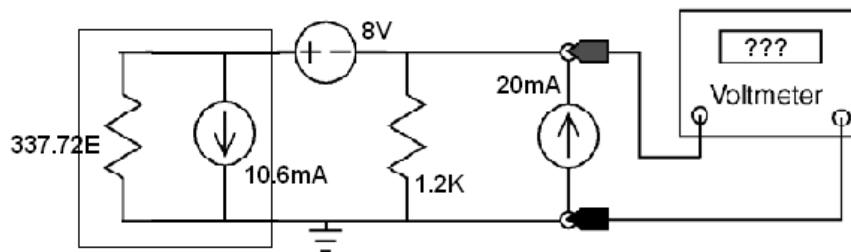
- Transform voltage source into current source.



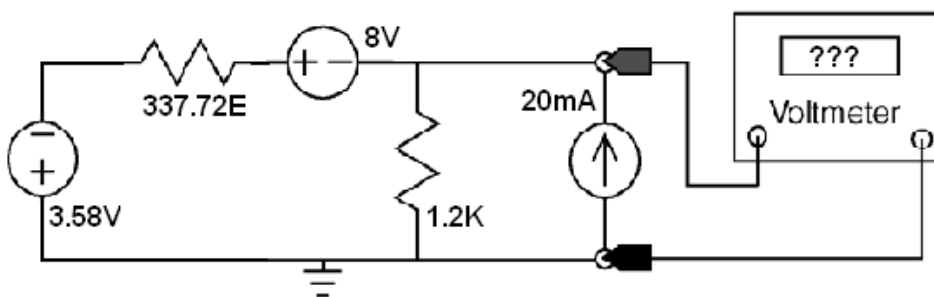
- Rearranging the parallel components,



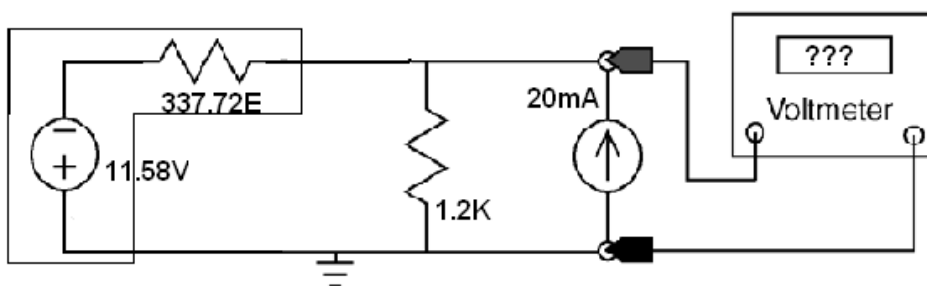
- Calculating the parallel resistance value



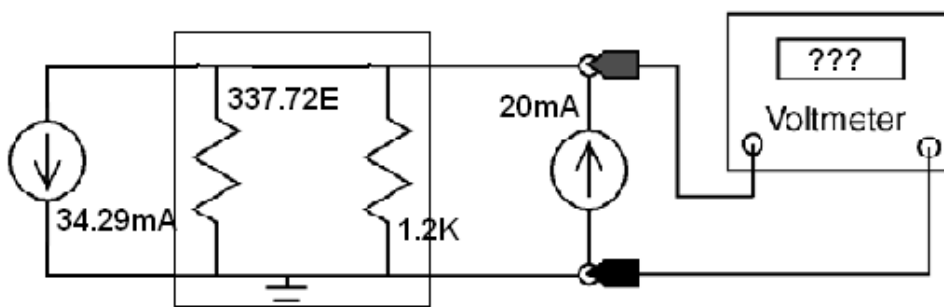
- Convert current source into voltage source.



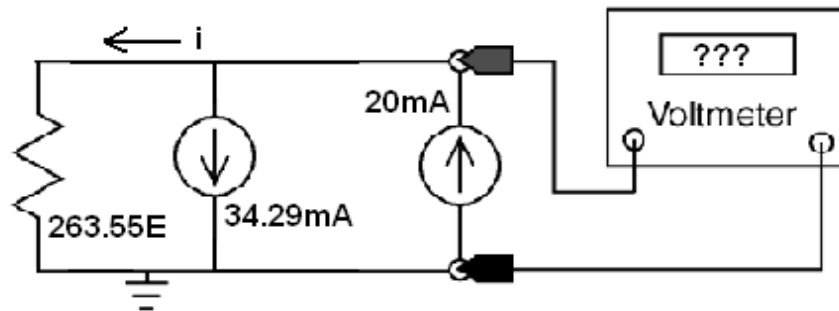
- Rearranging the series components and adding the voltage source in series.



- Convert voltage source into current source.



- Replace equivalent parallel resistance and rearrange the circuit.



Applying KCL at the top node of the circuit gives

$$i + 34.29\text{mA} = 20\text{mA}$$

$$i = -14.29\text{mA}$$

The voltage measured by the meter, is also the voltage across the resistor.

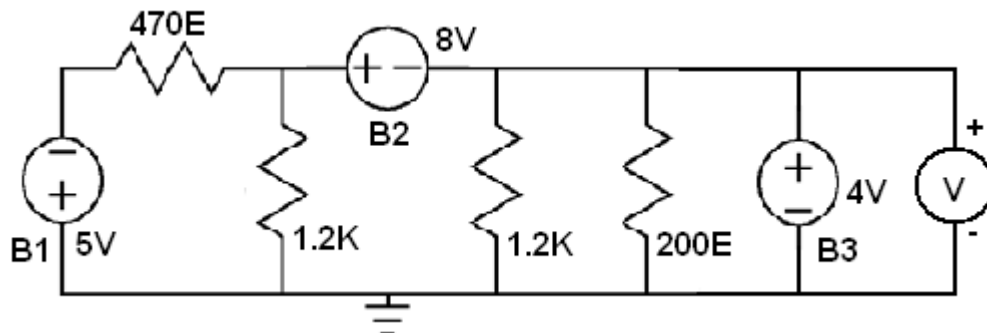
Ohm's law gives

$$V = -14.29 \times 263.55$$

$$V = -3.77\text{V}$$

Practical Verification of Mesh Current Analysis

- Do the connection as shown in circuit below.



- Set B1 = 5V, B2 = 8V and B2 = 4V and connect them into circuit with proper polarity.
- Switch on the power supply and measure the voltage V with voltmeter.
- Note down the readings and compare the theoretical and practical readings.

OBSERVATIONS:-

SR NO	Voltage (theoretical)	voltage (Practical)
1	V = -3.77V	V =

(Table given for reference only actual values may differ from above)

EXPERIMENT NO.13

EXPERIMENT NO. 13

NAME:-

Study of Ladder Network

OBJECTIVE:-

To study the resistor Ladder network and its application

EQUIPMENTS:-

- BEL-NBT Board
- Power supply
- Digital multimeter -1
- Voltmeters -3

THEORY:-

A resistor ladder is an electrical circuit made of repeating units of resistors. Two configurations are discussed below, a string resistor ladder and R-2R ladder.

An R-2R Ladder is a simple and inexpensive way to perform digital-to-analog conversion, using repetitive arrangements of precision resistor networks in a ladder-like configuration. A string resistor ladder implements the non-repetitive reference network.

String resistor ladder network (analog to digital conversion, or ADC)

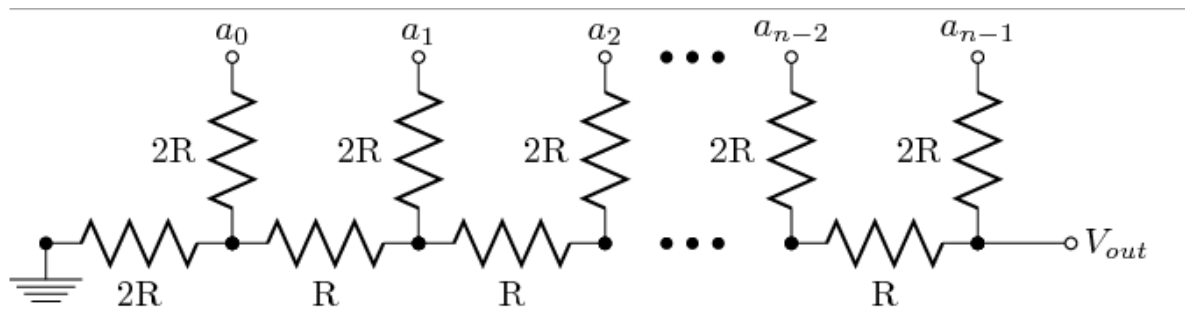
A string of many, often equally dimensioned, resistors connected between two reference voltages is a resistor string ladder network. The resistors act as voltage dividers between the referenced voltages. Each tap of the string generates a different voltage which can be compared with another *voltage*: this is the basic principle of a flash ADC (analog to digital converter). Often a voltage is converted to a current enabling the possibility to use R-2R ladder network.

- Disadvantage: for n-bits ADC the number of resistors grows exponentially, as $2^n - 1$ resistors are required, while the R-2R resistor ladder only increases linearly with the number of bits as it needs only $2n$ resistors.
- Advantage: Higher impedance values can be reached using the same number of components.

R-2R resistor ladder network (digital to analog conversion, or DAC)

A basic R-2R resistor ladder network is shown in Figure below. Bit4 MSB (most significant bit) to Bit0 LSB (least significant bit) are driven from digital logic gates. Ideally, the bits are switched between 0 volts (digital 0) and V_{ref} (digital 1). The R-2R network causes the digital bits to be weighted in their contribution to the output voltage V_{out} . In this circuit 5 bits are shown, giving 32 possible outputs. Depending on which bits are set to 1 and which to

0 the output voltage (*out*) will be a stepped value between 0 volts and (V_{ref} minus the value of the minimum step, Bit0). The actual value of V_{ref} (and 0 volts) will depend on the type (technology) of the digital logic gates used to drive Bit4-0



For a digital value VAL, of a R-2R DAC of N bits of 0 V/ V_{ref} , the output voltage V_{out} is:

$$V_{out} = V_{ref} \times VAL / 2^N$$

In the example shown, $N = 5$ and hence $2^N = 32$. With $V_{ref} = 3.3$ V (typical CMOS logic 1 voltage), V_{out} will vary between 00000, VAL = 0 and 11111, VAL = 31.

Minimum (single step) VAL = 1, we have

$$V_{out} = 3.3 \times 1 / 32 = 0.1 \text{ volts}$$

Maximum output (11111 VAL = 31, we have

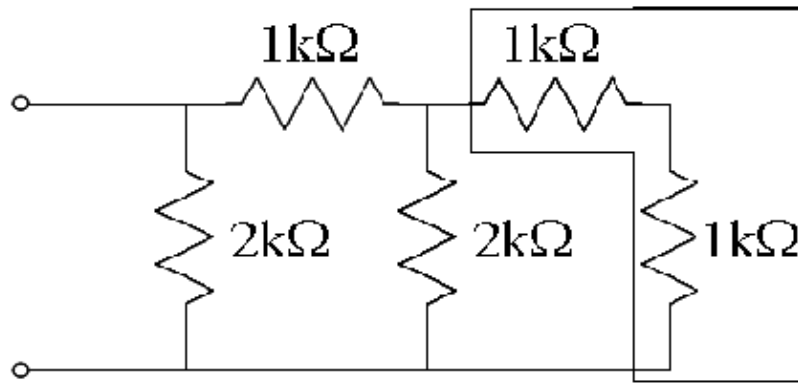
$$V_{out} = 3.3 \times 31 / 2^5 = 3.2 \text{ volts}$$

The R-2R ladder is inexpensive and relatively easy to manufacture since only two resistor values are required (or 1, if R is made by placing a pair of $2R$ in parallel, or if $2R$ is made by placing a pair of R in series). It is fast and has fixed output impedance R . The R-2R ladder operates as a string of current dividers whose output accuracy is solely dependent on how well each resistor is matched to the others. Small inaccuracies in the higher significant bit resistors can entirely overwhelm the contribution of the less significant bits. This may result in non-monotonic behavior at major crossings, such as from 01111 to 10000. Depending on the type of logic gates used and design of the logic circuits, there may be transitional voltage spikes at such major crossings even with perfect resistor values. These can be filtered, with capacitance at the output node for instance (the consequent reduction in bandwidth may be significant in some applications). Finally, the $2R$ resistance is in series with the digital output impedance. High output impedance gates (e.g., LVDS) may be unsuitable in some cases. For all of the above reasons (and doubtless others), this type of DAC tends to be restricted to a relatively small number of bits, although integrated circuits may push the number of bits to 14 or even more, the home constructor will typically limit themselves to 8 or less.

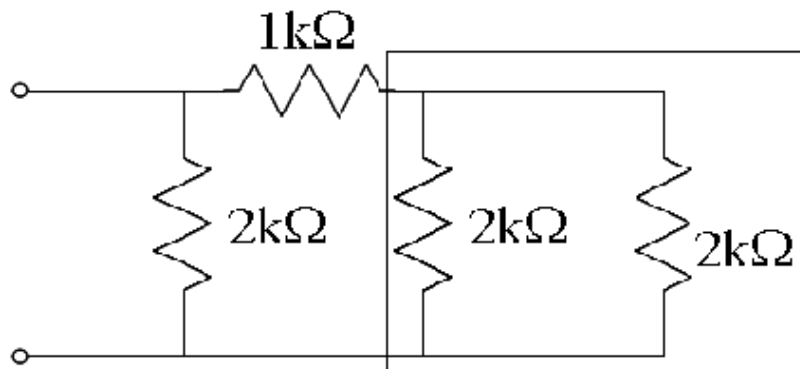
PROCEDURE:-

- Finding the equivalent resistance of the ladder network,

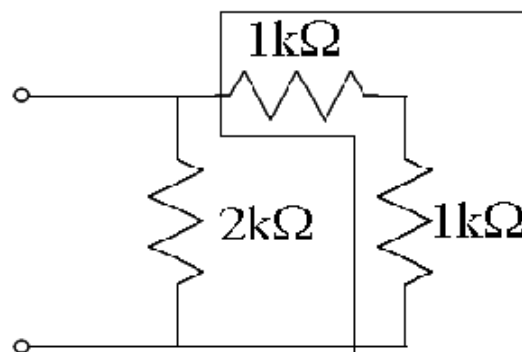
Theoretical analysis of finding equivalent resistance



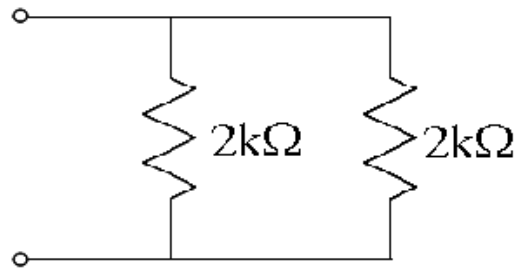
- Replace equivalent resistance of two series resistance of $1\text{k}\Omega$



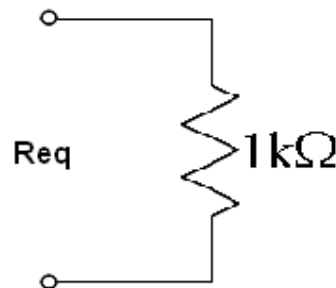
- Replace equivalent resistance of two parallel resistance of $2\text{k}\Omega$



- Replace equivalent resistance of two series resistance of $1\text{k}\Omega$

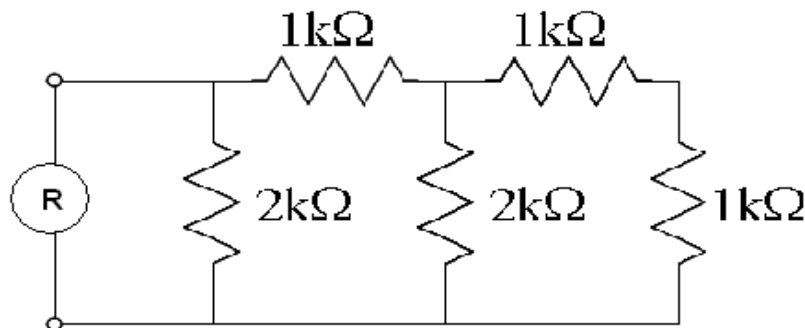


- Finally calculate the equivalent resistance of two parallel resistances of $2K$.



Practical Verification of Resistor Ladder Analysis

- Do the connection as shown in circuit below.



- Measure the equivalent resistance of resistance ladder R with multimeter.
- Note down the readings and compare the theoretical and practical readings.

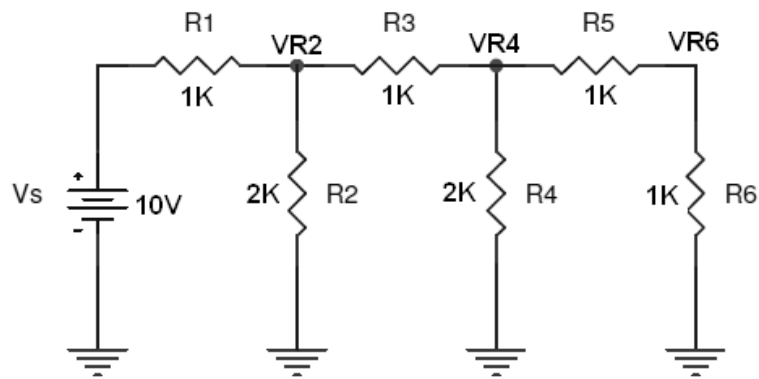
OBSERVATIONS:-

SR NO	Resistance (theoretical)	Resistance (Practical)
1	$R = 1K$	$R = 0.998K$

(Table given for reference only actual values may differ from above)

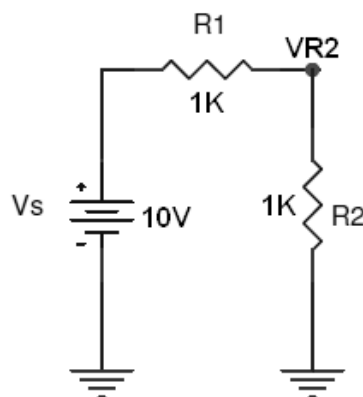
Finding the voltages derived by resistor ladder:

R-2R resistor networks (also called ladders) have a very unique property that divides the source voltage by 2 for each rung of the ladder that is crossed. R-2R ladders are often used when converting digital signals to analog and analog signals to digital. Because only two resistor values are needed the circuits can be made more accurately than some other circuits.

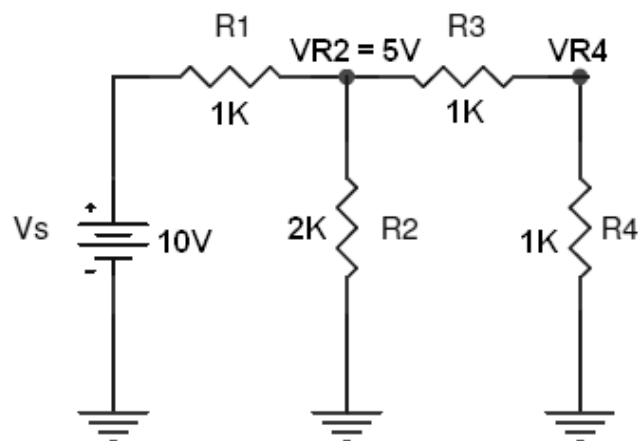


- To find voltage at R2 i.e. VR2, we need to calculate equivalent resistance at R2 due to R2, R3, R4, R5 and R6 using procedure A. the equivalent resistance at R2 will be 1K.
- Now apply voltage divider rule to find out VR2. See figure below.

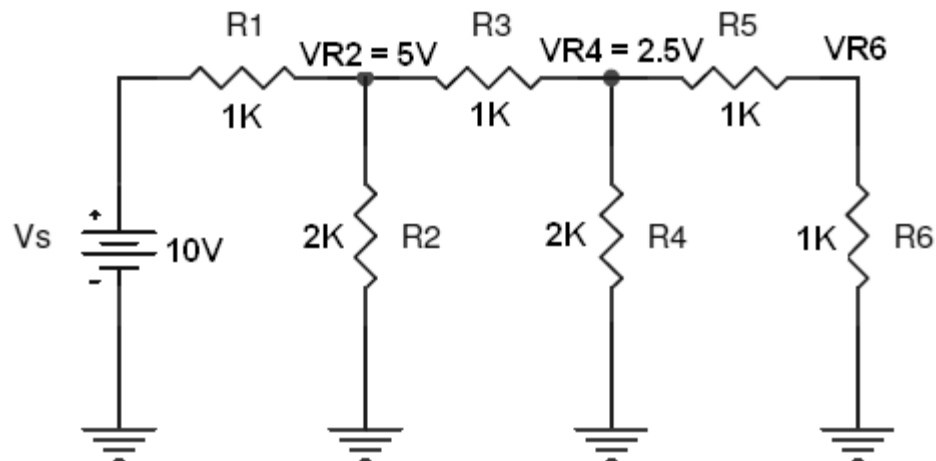
$$\mathbf{VR2 = 5V.}$$



- Similarly find VR4. **VR4 = 2.5V.** See fig below.

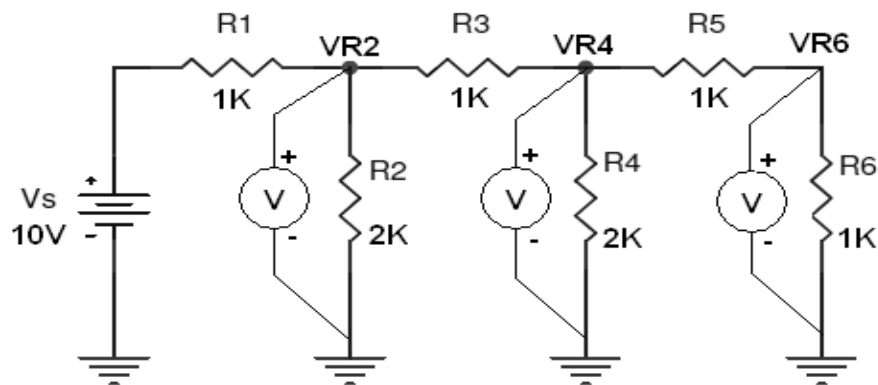


- Finally calculate the VR6. **VR6 = 1.25V**



Practical Verification Of Voltage Derived By Resistor Ladder

- Do the connection as shown in circuit below.



- Set $V_s = 10V$ and connect it with proper polarity to the circuit
- Switch on the power supply and Measure the voltages at VR2, VR4 and VR6 with respective voltmeter.
- Note down the readings and compare the theoretical and practical readings.

OBSERVATIONS:-

SR NO	Voltage (theoretical)	Voltage (Practical)
1	VR2 = 5V	VR2 = 5.01V
2	VR4 = 2.5V	VR4 = 2.502V
3	VR6 = 1.25V	VR6 = 1.250V

(Table given for reference only actual values may differ from above)

EXPERIMENT NO.14

EXPERIMENT NO. 14

NAME:-

Study of Millman's Theorem

OBJECTIVE:-

To study and verify millman's theorem

EQUIPMENTS:-

- BEL-NBT Board
- Power supply
- Ammeters - 3
- Voltmeter - 1

THEORY:-

Millman's Theorem states that for a circuit that can be redrawn as a network of parallel branches, with each branch consisting of a resistor or a resistor in series with a voltage source (or current source), the voltage V_m across all the parallel branches may be computed as follows: $V_m = (V_1/R_1 + V_2/R_2 + V_3/R_3 + \dots + V_n/R_n) / (1/R_1 + 1/R_2 + 1/R_3 + \dots + 1/R_n)$. For computation purposes, the value of the voltage source is equal to zero in a branch that consists of nothing but a resistor.

It then follows that the entire network of parallel branches may be simplified into a circuit consisting of just a voltage source V_m in series with a single resistor R_m , where $R_m = 1/(1/R_1 + 1/R_2 + 1/R_3 + \dots + 1/R_n)$. The current I_m through this simple circuit is given by: $I_m = V_m / R_m$, or $I_m = V_1/R_1 + V_2/R_2 + V_3/R_3 + \dots + V_n/R_n$.

Millman's Theorem, therefore, simplifies the computation of the voltage across the parallel branches of a circuit. Note that Millman's Theorem is only applicable to circuits that can be redrawn as a network of parallel branches, with each branch consisting of a resistor or a resistor in series with a voltage source or current source.

Millman's Theorem may also be written in terms of the impedance Z or admittance Y ($Z = 1/Y$) through each branch. Thus,

$$V_m = (V_1Y_1 + V_2Y_2 + V_3Y_3 + \dots + V_nY_n) / (Y_1 + Y_2 + Y_3 + \dots + Y_n);$$

$$Z_m = 1 / (Y_1 + Y_2 + Y_3 + \dots + Y_n); \text{ and}$$

$$I_m = V_1Y_1 + V_2Y_2 + V_3Y_3 + \dots + V_nY_n.$$

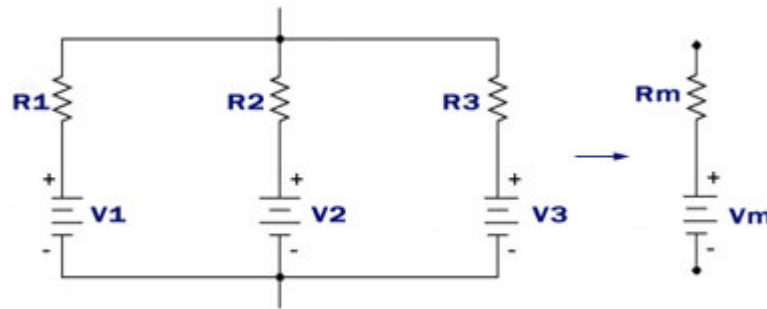
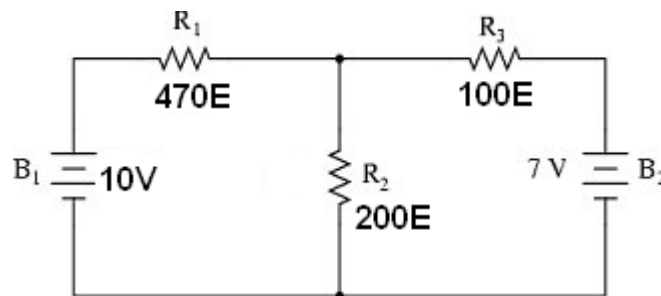


Figure illustrates Millman's theorem. In summary, Millman's Theorem is just a tool used for the quick computation of the voltage across and the current through a circuit that consists of nothing but parallel branches of a resistor in series with a voltage source.

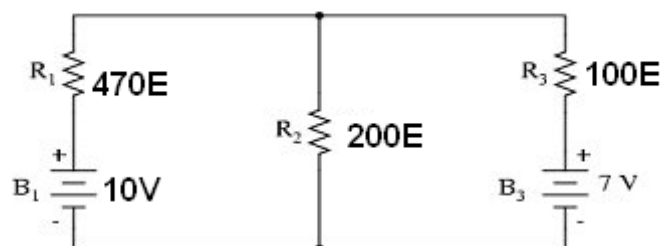
PROCEDURE:-

Theoretical Analysis Of Millman's Theorem

In Millman's Theorem, the circuit is re-drawn as a parallel network of branches, each branch containing a resistor or series battery/resistor combination. Millman's Theorem is applicable only to those circuits which can be re-drawn accordingly. Millman's theorem gives a very quick method to compute the voltage at the ends of a circuit made up of only branches in parallel.



When re-drawn for the sake of applying Millman's Theorem the circuit becomes



By considering the supply voltage within each branch and the resistance within each branch, Millman's Theorem will tell us the voltage across all branches. Please note that we've labeled the battery in the rightmost branch as "B3" to clearly denote it as being in the third branch, even though there is no "B2" in the circuit.

Millman's Theorem is nothing more than a long equation, applied to any circuit drawn as a set of parallel-connected branches, each branch with its own voltage source and series resistance:

Millman's Theorem Equation

$$\frac{\frac{E_{B1}}{R_1} + \frac{E_{B2}}{R_2} + \frac{E_{B3}}{R_3}}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}} = \text{Voltage across all branches}$$

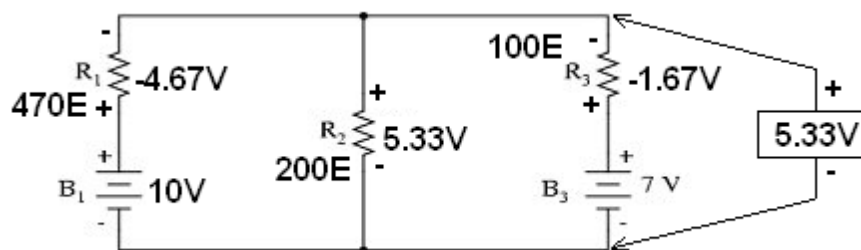
Substituting actual voltage and resistance figures from our example circuit for the variable terms of this equation, we get the following expression:

$$(10 / 470) + (0/200) + (7/100)$$

$$(1/ 470) + (1/200) + (1/100)$$

$$= 5.33V$$

The final answer of 5.33 volts is the voltage seen across all parallel branches like this.



The polarity of all voltages in Millman's Theorem are referenced to the same point. In the example circuit above, bottom wire of the parallel circuit is taken reference point, and so the voltages within each branch (10 for the R1 branch, 0 for the R2 branch, and 7 for the R3 branch) were inserted into the equation as positive numbers. Likewise, when the answer came out to 5.33 volts (positive), this meant that the top wire of the circuit was positive with respect to the bottom wire (the original point of reference). If both batteries had been connected backwards (negative ends up and positive ends down), the voltage for branch 1 would have been entered into the equation as a -10volts, the voltage for branch 3 as -7 volts, and the resulting answer of -5.33 volts would have told us that the top wire was negative with respect to the bottom wire (our initial point of reference).

With the result we get we can calculate the resistor voltage drops and branch currents like this:

$$E_{R1} = 5.33V - 10V = -4.67V \text{ (negative on top)}$$

$$E_{R2} = 5.33V - 0V = 5.33V \text{ (positive on top)}$$

$$E_{R3} = 5.33V - 7V = -1.67V \text{ (negative on top)}$$

To solve for branch currents, each resistor voltage drop can be divided by its respective resistance ($I=E/R$):

$$IR_1 = 4.67V / 470 = 9.9mA$$

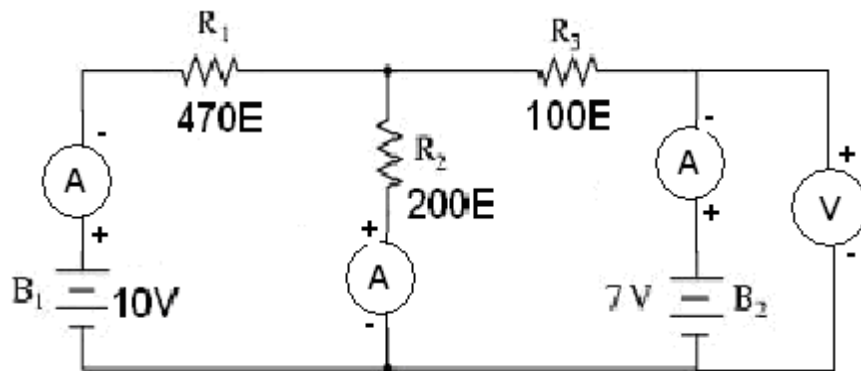
$$IR_2 = 5.33V / 200 = 26.65mA$$

$$IR_3 = 1.67V / 100 = 16.7mA$$

The direction of current through each resistor is determined by the polarity across each resistor, not by the polarity across each battery, as current can be forced backwards through a battery, as is the case with B3 in the example circuit. This is important to keep in mind, since Millman's Theorem doesn't provide as direct an indication of "wrong" current direction as does the Branch Current or Mesh Current methods. You must pay close attention to the polarities of resistor voltage drops as given by Kirchhoff's Voltage Law, determining direction of currents from that.

Practical verification of Millman's Theorem

- Do the connection as shown in circuit below.



- Set B1 = 10V, B2 = 7V and connect it with proper polarity to the circuit
- Switch on the power supply and Measure the voltage V across all the branches with voltmeter.
- Calculate voltage across R1, R2 and R3,
- Then calculate the Current Flowing through each branch.
- Note down the readings and compare the theoretical and practical readings.

OBSERVATIONS:-

SR NO	Voltage (theoretical)	Voltage (Practical)
1	V = 5.33V	V = 5.30V
SR NO	current (theoretical)	current (Practical)
1	IR1 = 9.9mA	IR1 = 9.3mA
2	IR2 = 26.65mA	IR2 = 24.3mA
3	IR3 = 16.7mA	IR3 = 14mA
SR NO	voltage(theoretical)	Voltage(Practical)
1	ER1 = -4.67V	ER1 = -4.73V
2	ER2 = 5.33V	ER2 = 5.29V
3	ER3 = -1.67V	ER3 = -1.636V

(Table given for reference only actual values may differ from above)

EXPERIMENT NO.15

EXPERIMENT NO. 15

NAME:-

Study of Kirchhoff's Law

OBJECTIVE:-

To study and verify KCL and KVL

EQUIPMENTS:-

- BEL-NBT Board
- Power supply
- Ammeters - 01
- Voltmeter - 01

THEORY:-

Kirchhoff's Voltage Law states that the algebraic sum of all the voltages around any closed path (loop or mesh) is zero.

Applying Kirchhoff's voltage law to the first and the second loops in the circuit shown in Figure 1 yields:

$$\text{Loop 1: } -V_s + V_1 + V_2 + V_5 = 0 \dots\dots\dots (1a)$$

$$\text{Loop 2: } -V_2 + V_3 + V_4 = 0 \dots\dots\dots (1b)$$

Kirchhoff's Current Law states that the algebraic sum of all the currents at any node is zero.

Applying Kirchhoff's current law to the first four nodes in the circuit shown in Figure1 yields the following equations;

$$\text{Node a: } -I_s + I_1 = 0 \dots\dots\dots (2a)$$

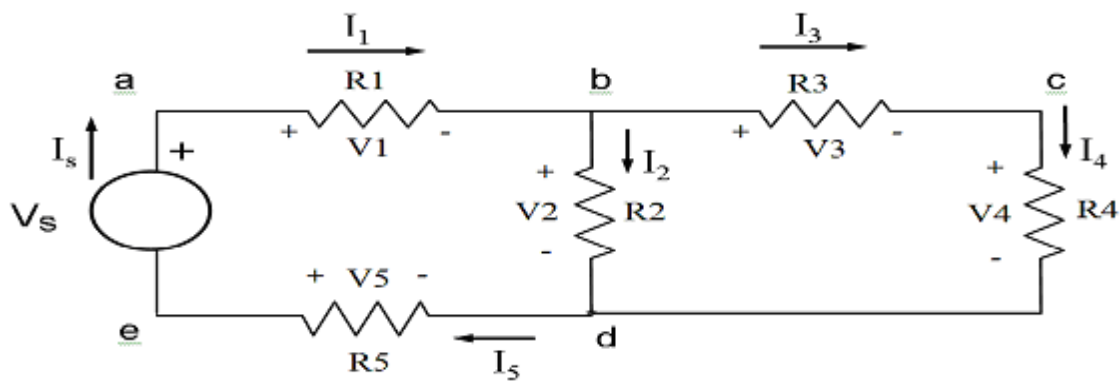
$$\text{Node b: } -I_1 + I_2 + I_3 = 0 \dots\dots\dots (2b)$$

$$\text{Node c: } -I_3 + I_4 = 0 \dots\dots\dots (2c)$$

$$\text{Node d: } -I_2 - I_4 + I_5 = 0 \dots\dots\dots (2d)$$

Sign Conventions

- All currents flowing towards a junction are assumed positive while all currents leaving a junction are assumed negative.
- The rise in potential is considered as positive and fall in potential is considered as negative.

CIRCUIT DIAGRAM:-**PROCEDURE:-**

- Construct the circuit shown in practical circuit diagram 1 using the values below:
- $R1 = 1\text{ K}$, $R2 = 2\text{ K}$, $R3 = 1\text{ K}$, $R4 = 5\text{ K}$, $R5 = 10\text{ K}$.
- Set the Variable Power Supply (Vs) to 5 Volts.
- Accurately measure all voltages and currents in the circuit one by one using the Digital Multi-Meter (DMM).
- Record the measurements in a tabular form containing the measured voltage and current values as shown below.
- Verify KVL for the loops in the circuit using equations 1a and 1b.
- Verify KCL for the nodes in the circuit using equations 2a, 2b, 2c and 2d.

OBSERVATION:-**For KVL Verification**

Sr. No.	Voltages	V (Volts)
1	VS	5
2	V1	0.402
3	V2	0.603
4	V3	0.099
5	V4	0.505
6	V5	3.988

(Table given for reference only actual values may differ from above)

For KCL Verification

Sr. No.	Current	I (ma)
1	IS	0.4
2	I1	0.4
3	I2	0.3
4	I3	0.09
5	I4	0.09
6	I5	0.4

(Table given for reference only actual values may differ from above)

EXPERIMENT NO.16

EXPERIMENT NO.16

NAME:-

Study of Compensation Theorem

OBJECTIVE:-

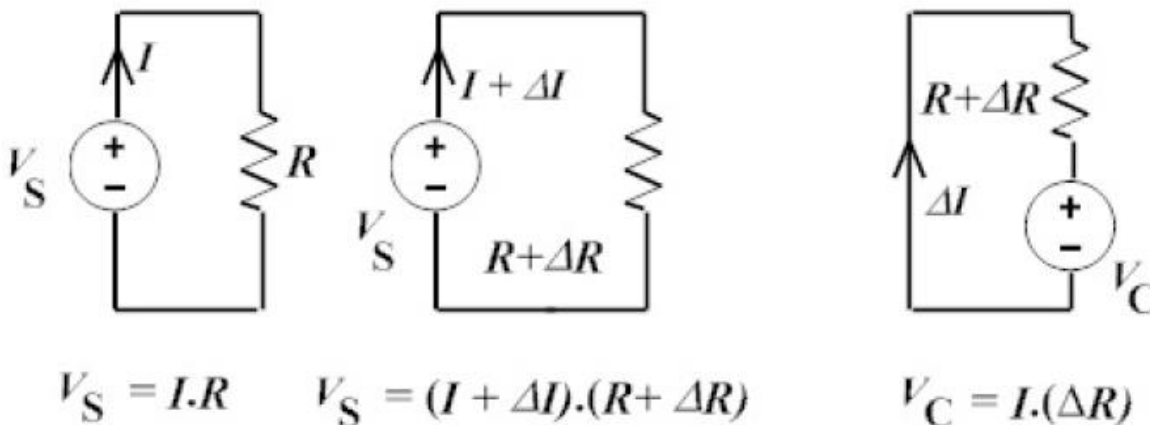
To study and verify Compensation theorem

EQUIPMENTS:-

- BEL-NBT Board
- Power supply
- Ammeters - 01

THEORY:-

Compensation theorem is a very useful theorem. It reflects the changes that can occur due to incremental changes in the value of a component. This theorem is illustrated by the circuit in Fig. below. Let the resistance change by a small amount. Its effect can be computed as shown. The effect can be computed by the response due to a voltage source V_C , the value of which is obtained as shown below.



$$V_S = I \cdot R = (I + \Delta I) \cdot (R + \Delta R) \quad (1)$$

$$\therefore V_S = I \cdot R, (I \cdot \Delta R) + \Delta I \cdot (R + \Delta R) = 0 \quad (2)$$

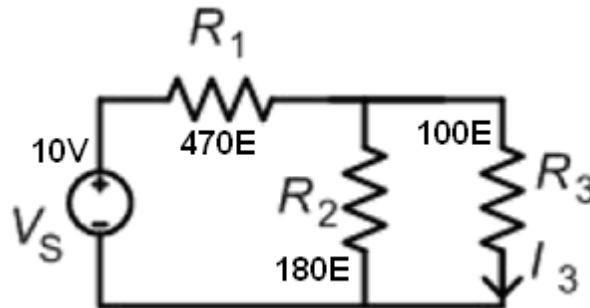
$$V_C = (I \cdot \Delta R), \Delta I = -\frac{V_C}{(R + \Delta R)} \quad (3)$$

From equation (1), cancel the common term on both sides of equation. Then we

Obtain equation (2). The value of compensating voltage V_C is obtained as shown by equation (3). The change in current is also obtained.

PROCEDURE:-

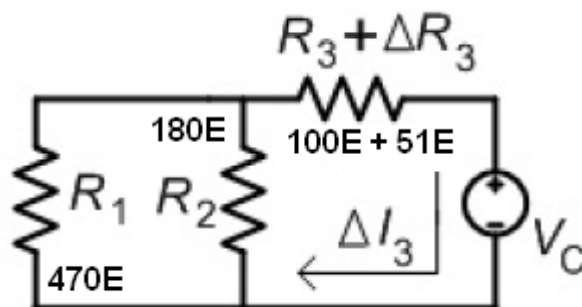
Theoretical Analysis of Compensation theorem,



Compensation theorem is illustrated by an example. For the circuit in Fig. above, obtain current I_3 .

$$I_3 = V_S \cdot \frac{R_2}{R_1 \cdot R_2 + R_2 \cdot R_3 + R_1 \cdot R_3} = 0.0120A$$

Let the value of resistor R_3 change by a small value (51E). Compute the change in current I_3 .



$$V_C = (I_3 \cdot \Delta R_3)$$

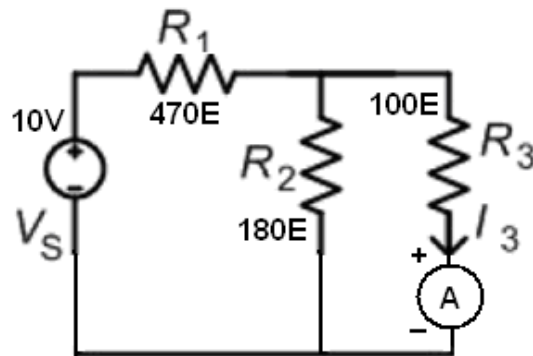
$$= (0.0120) (51) \\ = 0.612V$$

$$\Delta I_3 = - \frac{V_C}{(R_3 + \Delta R_3) + \frac{R_1 \cdot R_2}{R_1 + R_2}} = -0.00218A$$

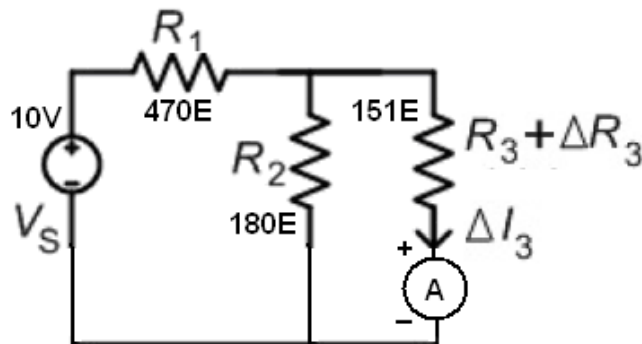
The values of components are specified. The value of current I_3 is obtained as shown. The value of compensating source V_C is then obtained and the change in Current I_3 , denoted by ΔI_3 , is obtained and its value has been shown above.

Practical verification of Millman's Theorem

- Do the connection as shown in circuit below.



- Measure the current I_3 from ammeter and note down the reading.
- Now replace R_3 by $151E$ resistor and again measure the current ΔI_3 .



- Calculate the change in current by taking difference between I_3 and ΔI_3 .
- Verify the theoretical and practical readings.

OBSERVATIONS:-

SR NO	current (theoretical)	current (Practical)
1	$\Delta I = 0.00218A$	$\Delta I = 0.002A$

(Table given for reference only actual values may differ from above)

EXPERIMENT NO.17

EXPERIMENT NO.17

NAME:-

Study of Maxwell's Bridge

OBJECTIVE:-

To study and verify Maxwell's Bridge

EQUIPMENTS:-

- BEL-NBT Board
- Power supply
- Voltmeter - 01

THEORY:-

The AC Bridge, a natural outgrowth of the DC bridge consists in its basic form of four bridge arms, a source of excitation, and a null-detector. The source of excitation is an AC signal at the desired frequency. The detector may be a set of an ac voltmeter, an oscilloscope, or another device capable of responding to alternating currents.

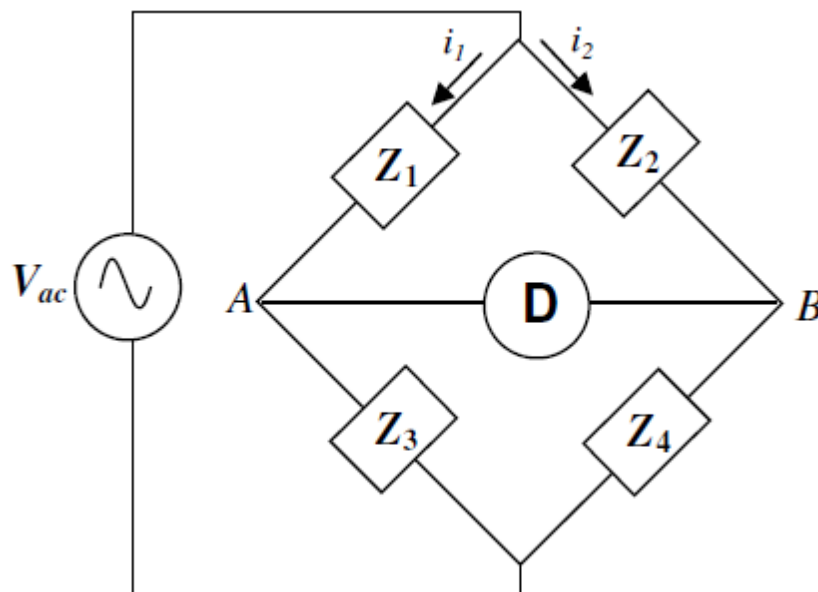


Figure 1 shows general AC Bridge.

The general form of an ac bridge is given in Fig.1. The four bridge arms Z_1 , Z_2 , Z_3 , and Z_4 are shown as unspecified impedances. The bridge is said to be balanced when the detector response is zero. One or more of the bridge arms are varied to balance the bridge so that a null response is obtained. The condition for bridge balance

requires that the potential difference from A to B be zero. This happens when the voltage at node A equals the voltage at node B , in both magnitude and phase.

In complex notation we can write

$$\begin{aligned} V_{Z_1} &= V_{Z_2} & ; & \quad i_1 Z_1 = i_2 Z_2 \\ V_{Z_3} &= V_{Z_4} & ; & \quad i_1 Z_3 = i_2 Z_4 \end{aligned} \quad (2.1)$$

Dividing the Equation,

$$\frac{Z_1}{Z_3} = \frac{Z_2}{Z_4} \quad (2.2)$$

$$Z_1 Z_4 = Z_2 Z_3 \quad (2.3)$$

If the impedances are written in polar form, Equation (2.3) can be rewritten as

$$(Z_1 \angle \theta_1)(Z_4 \angle \theta_4) = (Z_2 \angle \theta_2)(Z_3 \angle \theta_3) \quad (2.4)$$

To multiply these complex numbers we multiply the magnitudes and add the phase angles. Thus, equation (2.4) can be rewritten as

$$Z_1 Z_4 \angle (\theta_1 + \theta_4) = Z_2 Z_3 \angle (\theta_2 + \theta_3) \quad (2.5)$$

Equation (2.5) shows that two conditions must be met simultaneously when balancing an ac bridge.

The first condition is that the magnitude of the impedances satisfy the relationship

$$Z_1 Z_4 = Z_2 Z_3 \quad (2.6)$$

The second condition requires that the phase angles of the impedances satisfy the relationship

$$\angle \theta_1 + \angle \theta_4 = \angle \theta_2 + \angle \theta_3 \quad (2.7)$$

This expression states that the sums of the phase angles of the opposite arms must be equal. The Maxwell bridge shown in Fig. 2 measures an unknown inductance in terms of a known capacitance. Observing the bridge, we can see that

$$\begin{aligned} \therefore Z_1 &= \frac{R_1}{1 + j\omega R_1 C_1} & ; & \quad Z_2 = R_2 \\ \text{and} \quad Z_3 &= R_3 & ; & \quad Z_4 = R_x + j\omega L_x \end{aligned}$$

Substituting these expressions into Equation (2.6) and separating the real and imaginary terms yields

$$R_x = \frac{R_2 R_3}{R_1} \quad (2.8)$$

$$L_x = C_1 R_2 R_3 \quad (2.9)$$

Both Equations (2.8) and (2.9) must be satisfied for the bridge to be balanced.

PROCEDURE:-

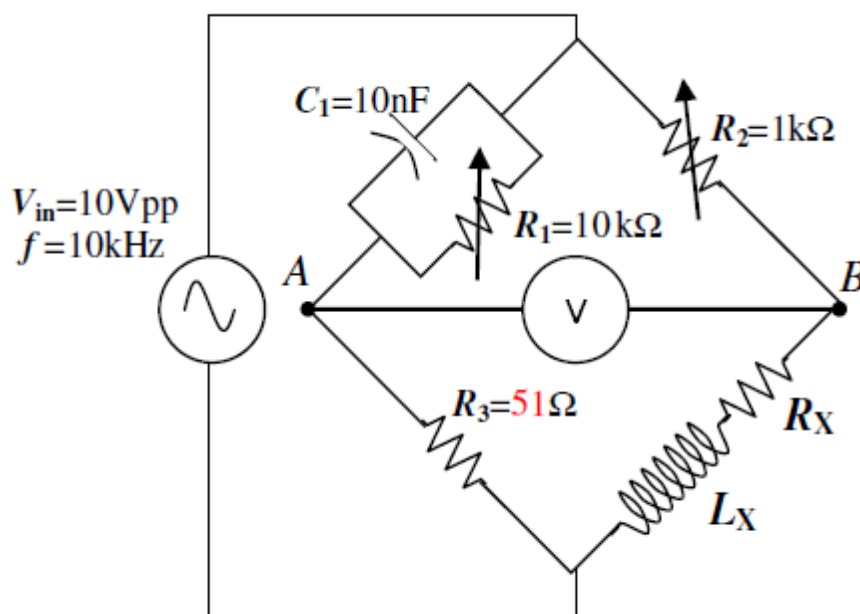


FIGURE 2 experimental circuit of Maxwell Bridge.

- Do the connection as shown in Figure 2.
- Connect $L_x = 1\text{mH}$ and $R_x = 100\Omega$.
- Set function generator sine wave output to 10 KHz, 10Vp-p and connect it to the bridge circuit.
- Connect voltmeter between A and b point and set it to AC mode.
- Balance the bridge by varying both R_1 and R_2 . Bridge is balanced when voltmeter shows almost 0V on its scale.
- Remove R_1 and R_2 from the circuit and measure and record their values. Calculate the unknown values of R_x and L_x using equation 2.8 and 2.9 respectively.
- Verify the theoretical and practical values.
- Repeat the experiment for different set of C_x and L_x .

OBSERVATIONS:-

Sr. No	Rx (Theoretical)	Lx (Theoretical)	Rx (Practical)	Lx (Practical)
1	100E	1mH	119E	0.51mH
2	200E	22uH	180E	38uH
3	180E	25uH	189E	37uH

(Table given for reference only actual values may differ from above)

ASSIGNMENT NO. 1

ASSIGNMENT NO.1

NAME:-

Study of Hay's Bridge

OBJECTIVE:-

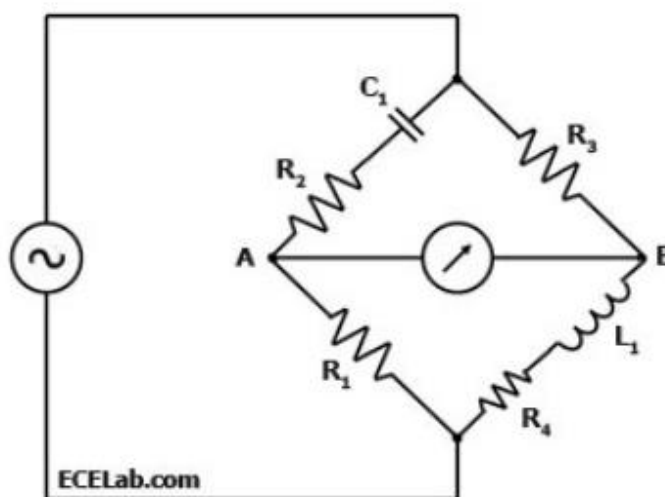
To study and verify Hays Bridge

EQUIPMENTS:-

- BEL-NBT Board
- Power supply
- Voltmeter - 01

THEORY:-

A **Hay Bridge** is an AC bridge circuit used for measuring an unknown inductance by balancing the loads of its four arms, one of which contains the unknown inductance. One of the arms of a Hay Bridge has a capacitor of known characteristics, which is the principal component used for determining the unknown inductance value. Figure 1 below shows a diagram of the Hay Bridge.



As shown in Figure above , one arm of the Hay bridge consists of a capacitor in series with a resistor (C_1 and R_2) and another arm consists of an inductor L_1 in series with a resistor (L_1 and R_4). The other two arms simply contain a resistor each (R_1 and R_3). The values of R_1 and R_3 are known, and R_2 and C_1 are both adjustable. The unknown values are those of L_1 and R_4 .

Like other bridge circuits, the measuring ability of a Hay Bridge depends on 'balancing' the circuit. Balancing the circuit in Figure 1 means adjusting R_2 and C_1 until the current

through the ammeter between points A and B becomes zero. This happens when the voltages at points A and B are equal. When the Hay Bridge is balanced, it follows that $Z_1/R_1 = R_3/Z_2$ wherein Z_1 is the impedance of the arm containing C_1 and R_2 while Z_2 is the impedance of the arm containing L_1 and R_4 . Thus, $Z_1 = R_2 + 1/(2\pi fC_1)$ while $Z_2 = R_4 + 2\pi fL_1$.

Mathematically, when the bridge is balanced,

$$[R_2 + 1/(2\pi fC_1)] / R_1 = R_3 / [R_4 + 2\pi fL_1]; \text{ or}$$

$$[R_4 + 2\pi fL_1] = R_3R_1 / [R_2 + 1/(2\pi fC_1)]; \text{ or}$$

$$R_3R_1 = R_2R_4 + 2\pi fL_1R_2 + R_4/2\pi fC_1 + L_1/C_1.$$

When the bridge is balanced, the reactive components are equal, so

$$2\pi fL_1R_2 = R_4/2\pi fC_1, \text{ or } R_4 = (2\pi f)^2 L_1R_2C_1.$$

Substituting R_4 , one comes up with the following equation:

$$R_3R_1 = (R_2 + 1/2\pi fC_1)((2\pi f)^2 L_1R_2C_1) + 2\pi fL_1R_2 + L_1/C_1; \text{ or}$$

$$L_1 = R_3R_1C_1 / (2\pi f)^2 R_2^2 C_1^2 + 4\pi fC_1R_2 + 1; \text{ or}$$

$L_1 = R_3R_1C_1 / [1 + (2\pi fR_2C_1)^2]$ after dropping the reactive components of the equation since the bridge is balanced.

Thus, the equations for L_1 and R_4 for the Hay Bridge in Figure 1 when it is balanced are:

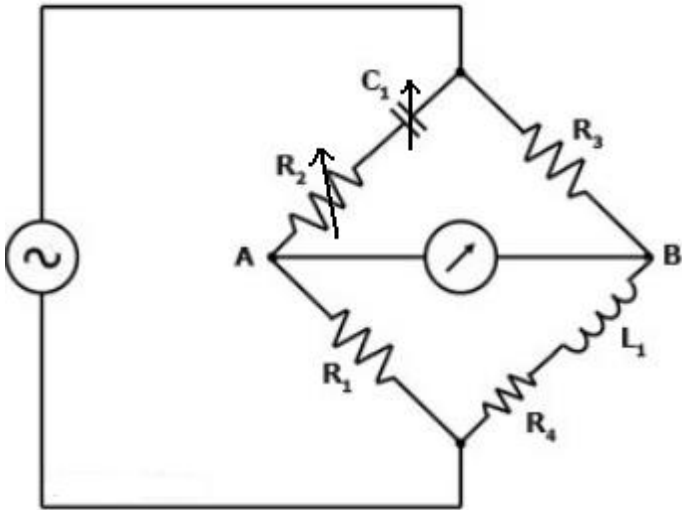
$$\mathbf{L_1 = R_3R_1C_1 / [1 + (2\pi fR_2C_1)^2];} \quad \mathbf{(1)}$$

$$\mathbf{R_4 = (2\pi fC_1)^2 R_2R_3R_1 / [1 + (2\pi fR_2C_1)^2]} \quad \mathbf{(2)}$$

Note that the balancing of a Hay Bridge is frequency-dependent.

PROCEDURE:-

- Refer the circuit below and equation 1 and 2 above and design the hays bridge components.
- Connect unknown inductor and resistor from banks provided.
- Connect voltmeter between A and B point and set it to AC mode.
- Balance the bridge by varying C_1 and R_2 .
- Once the bridge is balanced remove components C_1 and R_2 from the circuit, measure their values and calculate unknown inductor and resistor.
- Compare the theoretical and practical values.
- Repeat the procedure for another set of unknown components.



OBSERVATIONS:-

Sr. No	Rx (Theoretical)	Lx (Theoretical)	Rx (Practical)	Lx (Practical)
1				
2				
3				

ASSIGNMENT NO. 2

ASSIGNMENT NO. 2

NAME:-

Study of Schering Bridge

OBJECTIVE:-

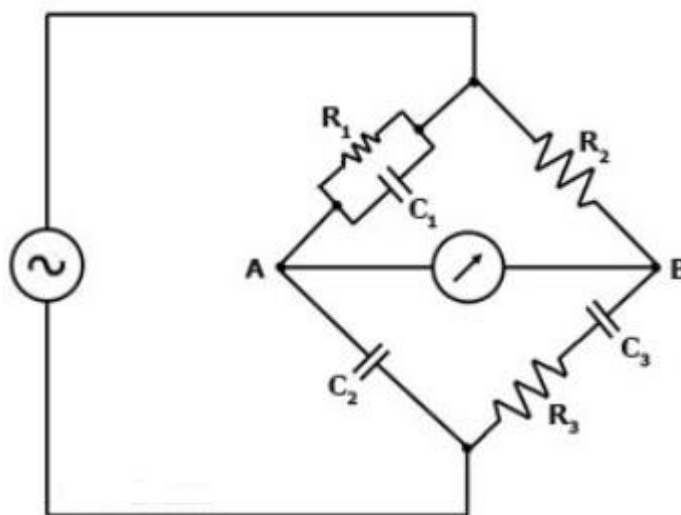
To study and verify Schering Bridge

EQUIPMENTS:-

- BEL-NBT Board
- Power supply
- Voltmeter - 01

THEORY:-

A **Schering Bridge** is a bridge circuit used for measuring an unknown electrical capacitance and its dissipation factor. The dissipation factor of a capacitor is the ratio of its resistance to its capacitive reactance. The Schering Bridge is basically a four-arm alternating-current (AC) bridge circuit whose measurement depends on balancing the loads on its arms. Figure 1 below shows a diagram of the Schering Bridge.



In the Schering Bridge above, the resistance values of resistors R_1 and R_2 are known, while the resistance value of resistor R_3 is unknown. The capacitance values of C_1 and C_2 are also known, while the capacitance of C_3 is the value being measured. To measure R_3 and C_3 , the values of C_2 and R_2 are fixed, while the values of R_1 and C_1 are adjusted until the current through the ammeter between points A and B becomes zero.

This happens when the voltages at points A and B are equal, in which case the bridge is said to be 'balanced'.

When the bridge is balanced, $Z_1/C_2 = R_2/Z_3$, where Z_1 is the impedance of R_1 in parallel with C_1 and Z_3 is the impedance of R_3 in series with C_3 . In an AC circuit that has a capacitor, the capacitor contributes a capacitive reactance to the impedance. The capacitive reactance of a capacitor C is $1/2\pi fC$.

As such, $Z_1 = R_1/[2\pi fC_1((1/2\pi fC_1) + R_1)] = R_1/(1 + 2\pi fC_1R_1)$ while $Z_3 = 1/2\pi fC_3 + R_3$.

Thus, when the bridge is balanced:

$$2\pi fC_2R_1/(1+2\pi fC_1R_1) = R_2/(1/2\pi fC_3 + R_3); \text{ or}$$

$$2\pi fC_2 (1/2\pi fC_3 + R_3) = (R_2/R_1)(1+2\pi fC_1R_1); \text{ or}$$

$$C_2/C_3 + 2\pi fC_2R_3 = R_2/R_1 + 2\pi fC_1R_2.$$

When the bridge is balanced, the negative and positive reactive components are equal and cancel out, so

$$2\pi fC_2R_3 = 2\pi fC_1R_2 \text{ or}$$

$$\mathbf{R_3 = C_1R_2 / C_2.} \quad \mathbf{(1)}$$

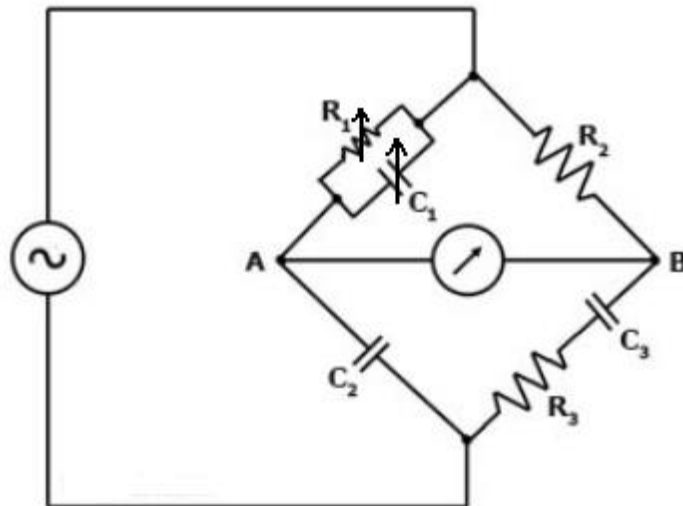
Similarly, when the bridge is balanced, the purely resistive components are equal, so $C_2/C_3 = R_2/R_1$ or

$$\mathbf{C_3 = R_1C_2 / R_2.} \quad \mathbf{(2)}$$

Note that the balancing of a Schering Bridge is independent of frequency.

PROCEDURE:-

- Refer the circuit below and equation 1 and 2 above and design the Schering bridge components.
- Connect unknown Resistor and Capacitor from banks provided.
- Connect voltmeter between A and B point and set it to AC mode.
- Balance the bridge by varying C_1 and R_1 .
- Once the bridge is balanced remove components C_1 and R_1 from the circuit, measure their values and calculate unknown Resistor and Capacitor.
- Compare the theoretical and practical values.
- Repeat the procedure for another set of unknown components.

**OBSERVATIONS:-**

Sr. No	Rx (Theoretical)	Cx (Theoretical)	Rx (Practical)	Cx (Practical)
1				
2				
3				

ASSIGNMENT NO. 3

ASSIGNMENT NO. 3

NAME:-

Study of Anderson Bridge

OBJECTIVE:-

To measure the self - inductance of a given coil by Anderson's bridge method

EQUIPMENTS:-

- BEL-NBT Board
- Power supply
- Function generator - 01
- Voltmeter - 01

THEORY:-

Anderson's bridge is the most accurate bridge used for the measurement of self – inductance over a wide range of values, from a few micro-Henries to several Henries. In this method the unknown self-inductance is measured in terms of known capacitance and resistances, by comparison. It is a modification of Maxwell's L – C bridge. In this bridge, double balance is obtained by the variation of resistances only, the value of capacitance being fixed.

Formula

Inductance of given coil $L = C [(R_1 + R_2) R_5 + R_2 R_4] \text{ mH}$

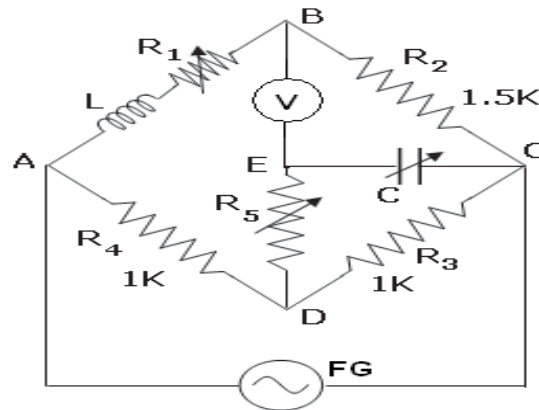
Where C = Capacity of the standard capacitor (uF)

R2, R3, R4 = Known, fixed and non – inductive resistances (KΩ)

R1, R5 = Variable resistances (KΩ)

PROCEDURE:-

- The circuit diagram of the bridge is as shown in the figure.



- The coil whose self-inductance is to be determined, is connected in the arm AB, in series with a variable non-inductive resistor R_1 . Arms BC, CD and DA contain fixed and non – inductive resistors R_2 , R_3 and R_4 respectively. Another non - inductive resistor R_5 is connected in series with a standard capacitor C and this combination is put in parallel with the arm CD.
- The head - phones are connected between B and E. The signal generator is Connected between A and C junctions.
- Select one capacitor and one inductor and connect them in appropriate places using patch chords. The Function generator frequency is adjusted to audible range.
- A perfect balance is obtained by adjusting R_1 and R_5 alternatively till the voltmeter shows 0V reading.
- The values of R_1 and R_5 are measured with a multi-meter(While measuring the R_1 and R_5 values, they should be in open circuit).In the balance condition
- The self – inductance value of the coil is calculated by using the above formula.
- The experiment is repeated with different values of C .

PRECAUTIONS:-

- 1) The product (CR_2R_4) must always be less than L .
- 2) R_1 and R_5 are adjusted until a close to 0V reading is observed on voltmeter.

OBSERVATIONS:-

SR NO	L (Theoretical)	L (Practical)
1		
2		
3		

ASSIGNMENT NO. 4

ASSIGNMENT NO. 4

NAME:-

Study of Owens Bridge

OBJECTIVE:-

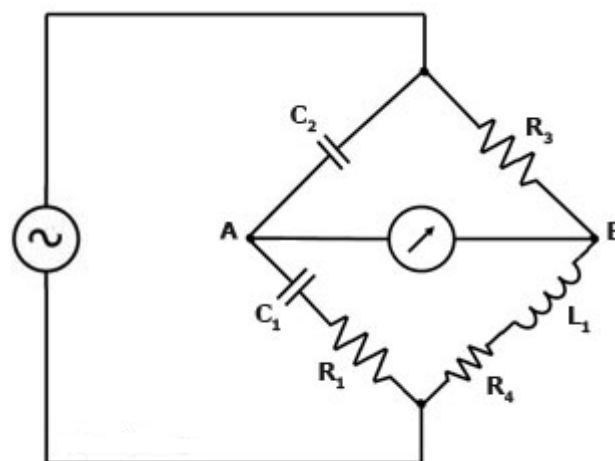
To study and verify Owens Bridge

EQUIPMENTS:-

- BEL-NBT Board
- Power supply
- Multimeter - 01

THEORY:-

An Owen Bridge is an AC bridge circuit used for measuring an unknown inductance by balancing the loads of its four arms, one of which contains the unknown inductance. Figure 1 below shows a diagram of the Owen Bridge.



The Owen Bridge

As shown in Figure, one arm of the Owen bridge consists of a capacitor in series with a resistor (C_1 and R_1) and another arm consists of an inductor L_1 in series with a resistor (L_1 and R_4). One arm contains just a capacitor (C_2) while the fourth arm just contains a resistor (R_3). The values of C_2 and R_3 are known, and R_1 and C_1 are both adjustable. The unknown values are those of L_1 and R_4 .

Like other bridge circuits, the measuring ability of an Owen Bridge depends on 'balancing' the circuit. Balancing the circuit in Figure means adjusting R_1 and C_1 until the current through the bridge between points A and B becomes zero. This happens when the voltages at points A and B are equal. When the Owen Bridge is

balanced, it follows that $Z_2/Z_1 = R_3/Z_4$ wherein Z_2 is the impedance of C_2 , Z_1 is the impedance of the arm containing C_1 and R_1 , and Z_4 is the impedance of the arm containing L_1 and R_4 . Mathematically, $Z_2 = 1/(2\pi f C_2)$; $Z_1 = R_1 + 1/(2\pi f C_1)$ while $Z_4 = R_4 + 2\pi f L_1$.

Thus, when the bridge is balanced,

$$\frac{1/(2\pi f C_2)}{[R_1 + 1/(2\pi f C_1)]} = R_3 / [R_4 + 2\pi f L_1]; \text{ or}$$

$$[R_4 + 2\pi f L_1] = (2\pi f C_2 R_3) [R_1 + 1/(2\pi f C_1)]; \text{ or}$$

$$R_4 + 2\pi f L_1 = 2\pi f C_2 R_3 R_1 + C_2 R_3 / C_1$$

When the bridge is balanced, the negative and positive reactive components are equal and cancel out, so
 $2\pi f L_1 = 2\pi f C_2 R_3 R_1$ or

$$L_1 = C_2 R_3 R_1.$$

Similarly, when the bridge is balanced, the purely resistive components are equal, so

$$R_4 = C_2 R_3 / C_1.$$

Note that the balancing of an Owen Bridge is independent of frequency.

PROCEDURE:-

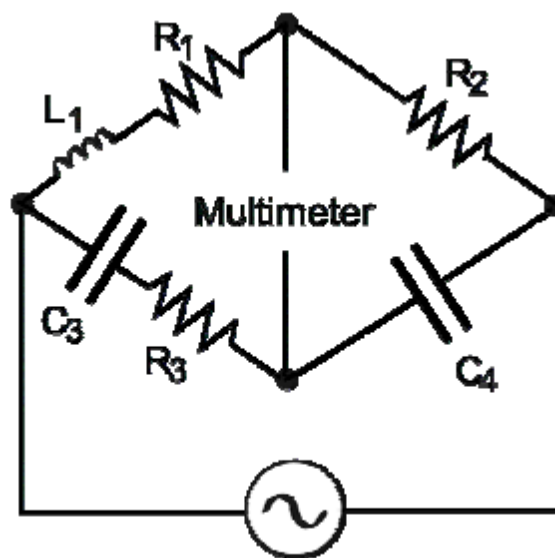


Figure 2

This circuit, shown in Figure 2, is suitable for the measurement of self-inductance.

- set up the circuit as in Figure 2 (check the wiring even if it appears to be set up correctly when you start).
- Using R_1 and R_3 as variables, C_3 and L_1 may be determined using different values of C_4 from 0.1 to 1.0 μF . Treat C_4 as a standard capacitor.
- A multimeter may be used as a detector. A frequency of 1 kHz is convenient. Students may have difficulty in finding a balance point first time.
- Start with $C_4 = 0.47 \mu\text{F}$ with C_3 nominally equal to the same value. From Equation

$$C_3 = C_4 (R_2 / R_1)$$

Equation 1

requires that R_1 and R_2 are then about the same. The value of the inductance is about 80-100 mH. If $L_1 = 0.09$ H then from equation

$$L_1 = C_4 R_2 R_3$$

Equation 2

If we assume that $R_2 = R_3 = R$,

$$0.09 = 0.5 \times 10^{-6} R^2$$

Giving $R = 424 \Omega$.

- Thus for $C_3 = C_4 = 0.47 \mu\text{F}$ one would expect a balance point
- When $R_1 = R_2 = R_3$. Find an approximate balance point with $R_1 = R_2 = R_3$.
- Then vary R_3 to satisfy equation (2) and vary R_1 to satisfy equation (3).
- Take a total of 10 measurements of L_1 and C_3 , for C_4 values from 0.1 to 1.0 μF

OBSERVATIONS:-

SR NO	C (Theoretical)	L (Theoretical)	C (Practical)	L (Practical)

ASSIGNMENT NO. 5

ASSIGNMENT NO.5

NAME:-

Study of LC Resonant

OBJECTIVE:-

To study LC resonant Circuits

EQUIPMENTS:-

- BEL-NBT Board
- Power supply

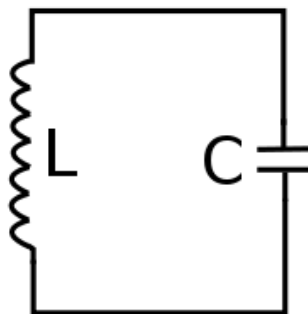
THEORY:-

An **LC circuit** is a **resonant circuit** or **tuned circuit** that consists of an inductor, represented by the letter L, and a capacitor, represented by the letter C. When connected together, an electric current can alternate between them at the circuit's resonant frequency.

LC circuits are used either for generating signals at a particular frequency, or picking out a signal at a particular frequency from a more complex signal. They are key components in many applications such as oscillators, filters, tuners and frequency mixers. An LC circuit is an idealized model since it assumes there is no dissipation of energy due to resistance. For a model incorporating resistance see RLC circuit.

OPERATION:-

An LC circuit can store electrical energy vibrating at its resonant frequency. A capacitor stores energy in the electric field between its plates, depending on the voltage across it, and an inductor stores energy in its magnetic field, depending on the current through it.



If a charged capacitor is connected across an inductor, charge will start to flow through the inductor, building up a magnetic field around it, and reducing the voltage on the capacitor. Eventually all the charge on the capacitor will be gone and the voltage across it will reach zero. However, the current will continue, because inductors resist changes in current, and

energy will begin to be extracted from the magnetic field to keep it flowing. The current will begin to charge the capacitor with a voltage of opposite polarity to its original charge. When the magnetic field is completely dissipated the current will stop and the charge will again be stored in the capacitor, with the opposite polarity as before. Then the cycle will begin again, with the current flowing in the opposite direction through the inductor.

The charge flows back and forth between the plates of the capacitor, through the inductor. The energy oscillates back and forth between the capacitor and the inductor until (if not replenished by power from an external circuit) internal resistance makes the oscillations die out. Its action, known mathematically as a harmonic oscillator, is similar to a pendulum swinging back and forth, or water sloshing back and forth in a tank. For this reason the circuit is also called a tank circuit. The oscillations are very fast, typically hundreds to billions of times per second.

Resonance Effect

The resonance effect occurs when inductive and capacitive reactance are equal in absolute value. (Notice that the LC circuit does not, by itself, resonate. The word resonance refers to a class of phenomena in which a small driving perturbation gives rise to a large effect in the system. The LC circuit must be driven, for example by an AC power supply, for resonance to occur (below).) The frequency at which this equality holds for the particular circuit is called the resonant frequency. The resonant frequency of the LC circuit is

$$\omega = \sqrt{\frac{1}{LC}}$$

where L is the inductance in Henries, and C is the capacitance in Farads. The angular frequency ω has units of radians per second.

The equivalent frequency in units of hertz is

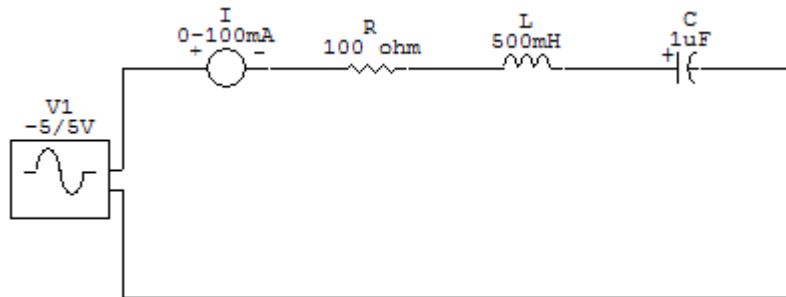
$$f = \frac{\omega}{2\pi} = \frac{1}{2\pi\sqrt{LC}}.$$

LC circuits are often used as filters; the L/C ratio is one of the factors that determines their "Q" and so selectivity. For a series resonant circuit with a given resistance, the higher the inductance and the lower the capacitance, the narrower the filter bandwidth. For a parallel resonant circuit the opposite applies. Positive feedback around the tuned circuit ("regeneration") can also increase selectivity (see Q multiplier and Regenerative circuit).

Stagger tuning can provide an acceptably wide audio bandwidth, yet good selectivity.

PROCEDURE:-

A) SERIES RESONANCE



- Connect the circuit as shown in figure above.
- Switch on the supply and adjust the signal generator to say 10V peak to peak.
- Vary frequency gradually from zero and note down the reading of Ac current I_0 in ammeter for different values of frequency.
- Take down the readings for smaller steps of frequency near the maximum current value to get accurate resonance frequency.
- From the tabulated readings plot a graph of current versus frequency. This graph is called as frequency response.
- From graph note down the resonant frequency (f_0) at which current is largest (I_0).
- Note down the frequency f_1 & f_2 corresponding to a current of $(I_0/\sqrt{2})$.
- Find Bandwidth and Q factor using the formula given below.
- f_1 = Lower half power frequency (Theoretical Equation $f_1 = f_0 - (R/4\pi L)$)
- f_2 = upper half power frequency (Theoretical Equation $f_2 = f_0 + (R/4\pi L)$)
- Band Width = $(f_2 - f_1)$ (Theoretical Equation $BW = R/2\pi L$ or $BW = f_0/Q$)
- Q Factor = $f_0 / (f_2 - f_1)$ (Theoretical Equation $Q_0 = (1/R) \cdot \sqrt{L/C}$)

DESIGN PROCEDURE:-

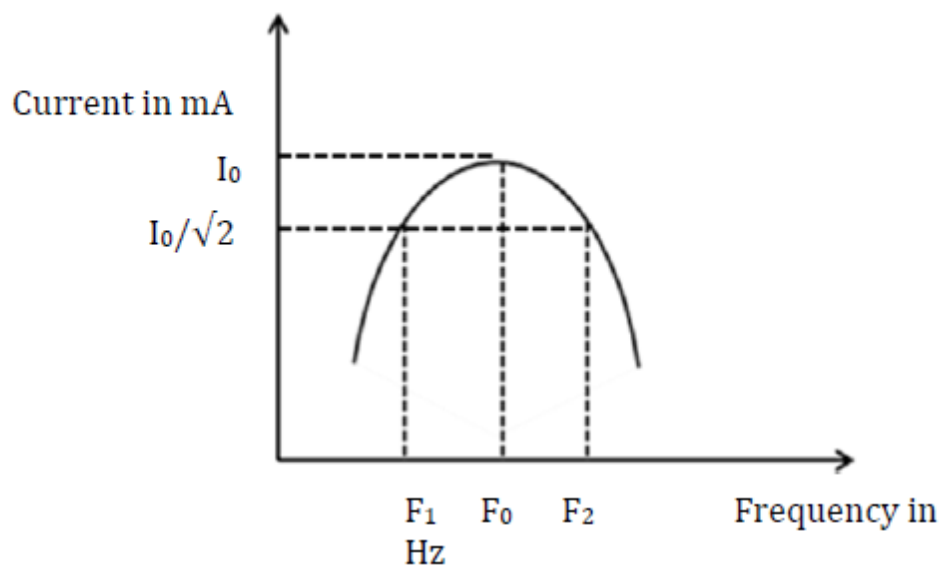
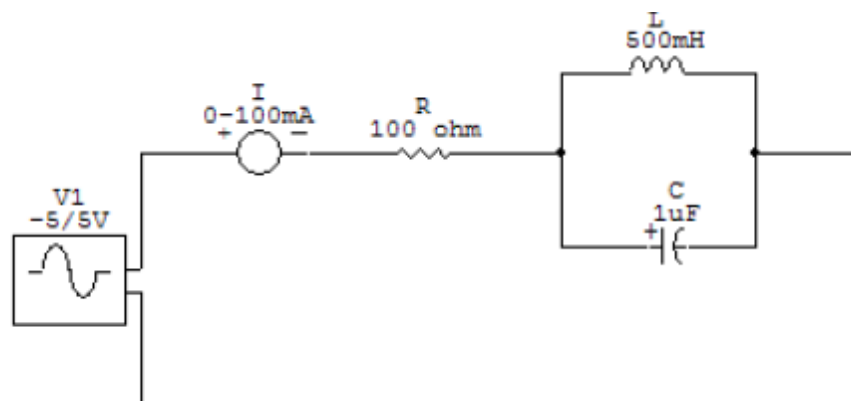
Assume $R=100\Omega$, $C=1\mu F$ and resonating frequency, $f_0 = 225\text{Hz}$ and calculate L.

Series Resonance: $f_0 = 1/2\pi\sqrt{LC}$

Parallel Resonance: $f_0 = 1/2\pi\sqrt{[(1/LC) - (R/2L)^2]}$

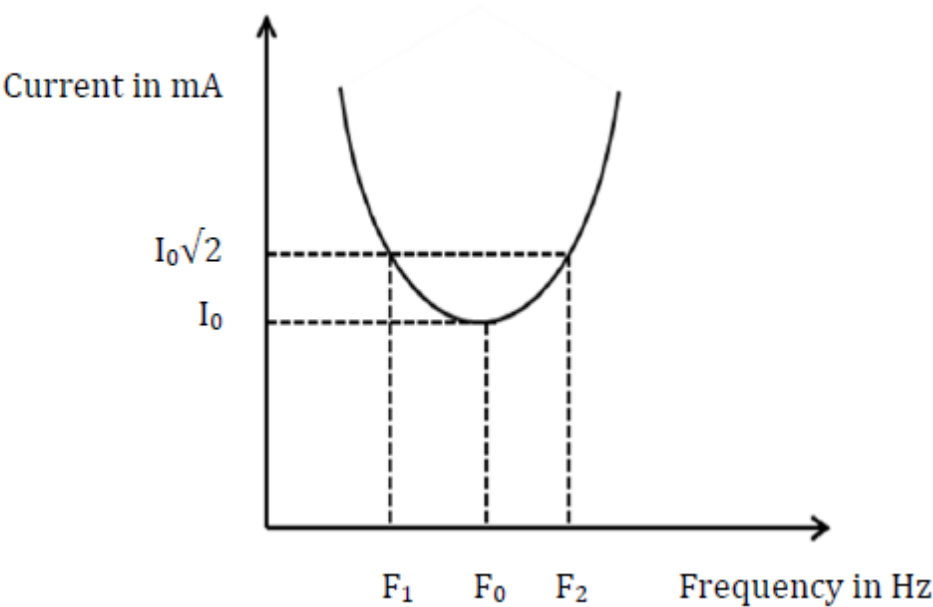
OBSERVATION:-

SR NO	F IN Hz	I IN mA
1		
2		
3		

GRAPH**B) Parallel Resonance****OBSERVATION:-**

SR NO	F IN Hz	I IN mA
1		
2		
3		

GRAPH



RESULT:-

Circuit Parameter	Series Resonance	Parallel Resonance
Theoretical resonating frequency		
Practical resonating frequency		
Theoretical bandwidth		
Practical bandwidth		
Theoretical Q Factor		
Practical Q Factor		

BASIC COMPONENT BANK

RESISTOR BANK						CAPACITOR BANK		INDUCTOR BANK	
R1	4.7E	R2	10E	R3	20E	C1	C2	POT1 1K	POT2 1K
R4	47E	R5	56E	R6	68E	33pF	47pF	POT3 10K	POT4 100K
R7	100E	R8	100E	R9	100E	C3	C4	POTENTIOMETER BANK	
R10	100E	R11	180E	R12	200E	100pF	150pF		
R13	200E	R14	200E	R15	220E	C5	C6		
R16	200E	R17	270E	R18	270E	330pF	10nF BOX		
R19	390E	R20	470E	R21	470E	C7	C8	L1 1mH	L2 3.3mH
R22	470E	R23	680E	R24	680E	0.1uF/50V	0.47uF/50V	L3 10mH	L4 100mH
R25	750E	R26	750E	R27	1K	C9	C10	L5 470uH	L6 10uH
R28	1K	R29	1K	R30	1K	1uF/50V	2.2uF/50V	L7 470uH	L8 10uH
R31	1K	R32	1.2K	R33	1.2K	C11	C12		
R34	1.5K	R35	1.5K	R36	2K	10uF/50V	47uF/50V		
R37	2K	R38	2K	R39	2.2K	Gc1 PF	Gc2 UF		
R40	2.2K	R41	4.7K	R42	4.7K				
R43	5.1K	R44	5.1K	R45	5.6K				
R46	5.6K	R47	6.8K	R48	10K				
R49	10K	R50	10K	R51	20K				
R52	100K	R53	120K	R54	150K				
BEL-NBT : NETWORK AND BRIDGES TRAINER									

BEL-NBT : NETWORK AND BRIDGES TRAINER